



PC-1

SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY- 2nd REVISION)

ORIGINAL APPROVED COST	PKR Million. 6,989.875/-
1st REVISED APPROVED COST	PKR Million. 14,165.060/-
2nd REVISED PROPOSED COST	PKR Million. 49,270.400/-
1st REVISED APPROVED GESTATION	30 Months Till May 2022
2nd REVISED PROPOSED GESTATION	91 Months Till June 2027
APPROVAL FORUM	ECNEC (ECNEC)

1. NAME OF THE PROJECT

SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCHLESS TECHNOLOGY- 2nd REVISION)

2. LOCATION OF THE PROJECT

2.1. DISTRICT(S)

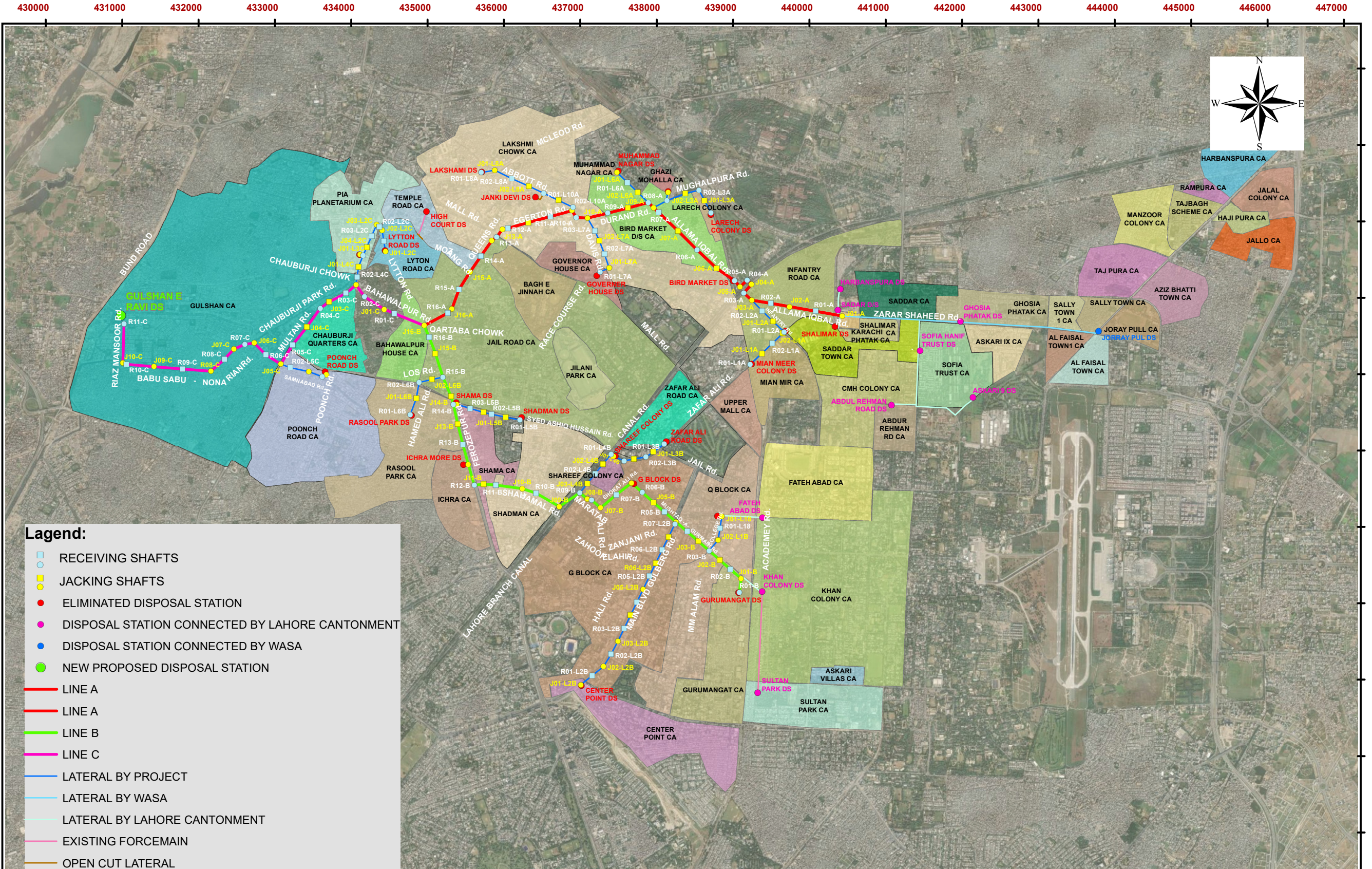
I. LAHORE

2.2. TEHSIL(S)

I. LAHORE CANTT., LAHORE CITY, SHALIMAR

2.3. CONSTITUENCY

I. PP-142, PP-147



LAHORE WATER & WASTE WATER MANAGEMENT PROJECT-SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THOURGH TRENCHLESS TECHNOLOGY)

3. AUTHORITIES RESPONSIBLE FOR

3.1. SPONSORING AGENCY

- GOVERNMENT OF PUNJAB

3.2. EXECUTION AGENCY

- WASA LDA

3.3. OPERATIONS AND MAINTENANCE AGENCY

- WASA LDA

3.4. CONCERNED FEDERAL MINISTRY

- PLANNING, DEVELOPMENT AND REFORM

4. PLAN PROVISION

Sr #	Description
1	Source of Funding: Scheme Listed in ADP CFY
2	GS No: 3036
3	Total Allocation: 29.500

Comments:

Instant scheme was approved from ECNEC for an amount of 14165.06 Million vide A.A. No. SO(UD)3-4/2019 dated 25.06.2020 (**Annexure-01**) based on working of M/s NESPAK in 2019. However, revision of PC-I is proposed owing to upward revision in cost worked out by PMC (JV of American Firm M/s EBA with EEC Global, ECSP & TEAM) owing to following reasons:

1. Increased design flow owing to updated route alignment.
2. Increase in trunk sewer depth, length, diameter, catchment areas & beneficiary population.
3. Updated working on MTBM Quantity since, insufficient MTBMs were taken in last PC-I to complete the assignment within stipulated time frame.
4. Missing essential components like intake structures, odour control, drop structures, ventilation system & Grid station etc.
5. Updated MRS & Non-MRS rates after soaring inflation period of five years from 2020-2024
6. Construction of New Disposal (848 cfs).

It is Pertinant to mention that instant scheme was cleared by PDWP in its 26th Meeting held on 09.09.2024 with cost of PKR 49,270.40 Million (**Annexure-02**)

5. PROJECT OBJECTIVES

Project Objectives

OBJECTIVES OF PROJECT

Instant project of Laying of Trunk Sewer through Trenchless Technology commences from Karachi Pathak Saddar Cantt and Gurumanget Gulberg to Gulshan-e-Ravi Disposal station and cover almost 15658.2 Acres Catchment area with serving population of about 2.35 Million People, while passing through Central Zone of Lahore. Sewerage system of Central Zone of Lahore that was installed about 30-40 years ago has passed its useful life and thus become inadequate to cope with increased demands of thickly populated urbanized areas in Central parts of City. Hence, main objective of the Trunk Sewer Installation Project is to provide improved facility of sewage disposal system in the Central Zone of Lahore (the project area) for proper collection, conveyance, and disposal of sewage effluent to the proposed disposal station in Gulshan-e-Ravi area by providing new trunk sewers and Contributory Allied connecting Sewers. The project will also facilitate and aid in redressing chronic sewerage related issues in Northern parts of Lahore City like Harbanspura, Tajpura, Fatehgarh areas etc., of Tehsil Shalimar and Cantt of Lahore city as Leg A of this project will cater flow originating from another ongoing ADP scheme of WASA pertaining to sewerage system in Harbanspura and adjoining areas. In addition to forgoing, allowance of sewage flows from Cantonment areas like Ghosia Phatak, Askari IX, Sofia Hanif, Abdur Rehman Road etc., has also been taken into consideration while formulating the instant scheme. Benefits of project are summarized hereunder

Implementation of this project will result in:

- Environmental Improvement.
- Restoration of Recreational activities along Cantonment Drain.
- Elimination of pumping requirement at the intermediate disposal stations, thus saving in higher Energy Costs cum Maintenance Costs and Manpower Requirements that use to be deployed earlier at disposal stations round the clock.
- Existing primary drains, which are presently being used as sullage carriers, will start acting as storm water channels, which can be perceived as primary purpose of the project.
- Mitigation of Storm Water Ponding issues in Beneficiary Areas in general and highly built-up deeper areas of Central zone like Qartaba Chowk, Lakshmi Chowk, ichra, Shama, Shadman, Samanabad more etc., in particular. Resultantly, it can be envisaged that all issues concerning to disruption in Livelihood Activities of residents living in suburbs of stated areas will also be addressed.
- Construction of Sewer through trenchless technology at depths ranging from 10.63 m to 17.28 m will aid in minimum disruption of Traffic flow thus avoiding

resettlement of people or re-construction of civic infrastructures thereof as it often faces in conventional open cut method, hence, it can be perceived that there will be meager impact on livelihood activities during execution of work.

- In addition to above WASA Lahore has planned to construct a wastewater treatment plant of about 178 MGD (2035) Ph-I future extension 225 MGD (2045) in Ph-II at Babu Sabu which profoundly will accommodate flows originated from Central Zone of Lahore city, hence, after reinstating the Cantonment drain into its initial condition as storm water carrier, it will be simpler to divert sewage outflows from Gulshan-Ravi Disposal to proposed treatment plant, thus avoiding further cost implications for separate conveyance system of this treatment plant without compromising on larger design capacity issues.

6. DESCRIPTION AND JUSTIFICATION OF PROJECT

6.1 JUSTIFICATION OF PROJECT:

Description & Justification

GENERAL DESCRIPTION

Lahore is the capital of Punjab province. It is an old city and has a past history of centuries. The city remained under the influence of Mughal Empire for quite a long period of time. It is a city of international importance and has important travel facilities like international Allama Iqbal Airport and the city is well connected with rest of Pakistan through extensive road and railway networks.

The old walled city of Lahore, having very narrow streets, is highly congested and almost all the important business activities exist in this area. The peripheral part of the city, especially the area close to River Ravi, is spreading out in a haphazard and disorganized way. However, areas like Defence Housing Colony, Gulberg, Cantonment, Bahria Town, Shadman, Model Town, Township, Samanabad and Gulshan-e-Ravi are the major well-planned and well-organized housing developments. Some of the important features of Lahore city are as follows:

AREA AND POPULATION

Lahore city covers a total land area of 404 km². Its limits are revised/ increased from time to time, keeping in view the growing demands and expansion trends. The population as per census 2017 is 11.13 million and is still growing at a rate of 2.83% per year. According to the Digital Census conducted in year 2023 by PBS, population of Lahore come out to be 13 Million.

LOCATION/TOPOGRAPHY

Lahore city is bounded on the north and west by the Sheikhpura District, on the east by Wagah, and on the south by Kasur District. The river Ravi flows on the north-western side of Lahore. Lahore district lies between 31°15' and 31°43' North latitudes and 74°10' and 74°39' East longitudes. The topography of Lahore district comprises the central uplands and alluvial lands of the river Ravi with average elevation of about 150 to 200 meters above mean sea level (MSL). Upland is a plain slope from north-east to south-west and all the area drains into River Ravi.

Two major industrial estates i.e., Quaid-e-Azam Industrial Estate and Sundar Industrial Estate are located in the vicinity of Lahore, which are producing many industrial goods, thus not only providing economic boost to the country, but also providing job opportunities to millions of people of Pakistan. In addition, many small industries also exist inside Lahore city, especially in the walled city part, leading to the country's uplift and providing job opportunities to the local communities.

EXISTING WASTEWATER INFRASTRUCTURE

The existing WASA service area for sewerage system covers about 350 km². Based on topographical features and existing sewerage network, WASA has divided this service area into the following six (6) sewerage catchment Zones.

- Central Zone;
- South Zone;
- South East Zone;
- Mehmood Booti Zone;
- Khokhar Road Zone; and
- Shahdara Zone

Figure 1 shown below exhibits the boundaries of these zones. Sewerage and drainage system of these zones is also shown in this figure. As per data available with WASA, the existing sewerage system consists of 4,022 km length of trunk and lateral sewers, 16 major sewage disposal pumping stations and 112 intermediate lift stations. The whole wastewater is ultimately discharged into River Ravi without any treatment. Some of it is being used for agricultural purposes as well.

The current project covers the Central Zone of Lahore only, which has an area of 100.26 square kilometers. The details of wastewater collection system in the Central Zone catchment area are discussed in the following sections.

EXISTING SEWERAGE SYSTEM IN CENTRAL CATCHMENT ZONE

The existing sewerage system of Central zone of Lahore (the project area) was laid during the past 25 to 60 years period and the collected sewage effluent is being pumped through disposal/lift stations into the open storm water drains which traverse through the heart of the city, causing severe environmental degradation and odor problems. Moreover, during monsoon season, due to inadequate capacity in the storm water channels, the rainfall accumulates at low points causing flooding in many places and results in further degradation of sanitary environment.

The sewerage system in the Central Catchment Zone consists of 233 km of existing trunk sewers and 1,092 km of lateral sewers, making a total length of 1,325 km. There are four (4) main drains, six (6) major sewage pumping stations, two (2) drainage pumping stations and twenty-seven (27) intermediate lift stations. The intermediate lift and pumping stations terminate into five (5) major existing disposal stations. The effluent from these disposal stations is pumped into disposal channels connected to the Ravi River.

The existing sewerage system mainly consists of reinforced concrete sewer pipes while in some areas older brick masonry sewers also exist. Salient features of the existing final disposal stations near the Ravi River are shown in **Table 1**

Table 1 : Salient Features of Existing Outfall Disposal Stations in Central Zone

Disposal Station	Location	Year of Commissioning	Total Capacity (cusec)	Total Number of Pumps
Bhati Gate	Near Data Darbar	2000	100	4
Main Outfall-1	Bund Road near Saggian Bridge	1945	181	10
Main Outfall-2	Bund Road near Saggian Bridge	1977	102	4
Main Outfall-3	Bund Road near Saggian Bridge	1985	87	5
Gulshan-e-Ravi	G-Block, Gulshan-e-Ravi near Bund Road	1992	560	14
Multan Road	Sabzazar Scheme, adjacent to Motorway Bypass	1982	240	6

STORM WATER DRAINS IN CENTRAL ZONE

The drainage network of Central Zone comprises mainly of Cantonment Drain and its tributaries. Total catchment area for Cantonment Drain has been estimated to be 23,653 acres. Some secondary/tertiary drains catering to specific areas of Central Zone are as below

- Cantonment Drain
- Central Drain (Main)
- Gulberg Drain No.1
- Gulberg Drain No.2
- Gulshan-e-Ravi Drain

The project's end goal is to eliminate the disposal of sewage discharge into the storm water drains and to enhance the capacity of the existing sewerage system. Drains that lie within the project area are discussed in the following:

Cantonment Drain

The total length of this drain starting from Jorray Pul to the New Babu Sabu pumping station is about 19.2 km. Its width varies from 3.7 m to 29 m, while its depth ranges from 2 m to 4.6 m. Its original section is generally reduced due to dumping of solid wastes and encroachments at several locations. As such the drain flows almost at full capacity even during the dry season.

Central Drain

The Central Drain is one of the main tributaries of Cantonment Drain serving the most critical areas of the Central Zone like Lakshami Chowk, Railway Station, Gawal Mandi, Macleod Road, Brandreth Road and Hall Road. The total length of this drain is about 4 km, and the width of the drain varies from 3.5 m to 9.5 m while the depth ranges from about 1.2 m to 3.8 m.

This drain is badly silted /filled up and encroached upon at several places. This has reduced its original cross-section considerably for storm water disposal. Due to the ingress of sewage from adjoining areas, this drain is now acting as a sullage carrier and it flows at almost its full capacity, like Cantonment Drain, even in the dry season.

Gulberg Drain No.1

Gulberg Drain No.1 is an earthen trapezoidal section drain with a high-tension electrical transmission line running parallel along the drain. A lot of dumping of solid waste was observed just after the canal crossing which obstructs the flow. This drain serves several areas including Gulberg and Cantt. areas. Starting from the Garrison Mess, it joins Gulberg Drain No.2 near the Gulberg Main Boulevard. It is mostly uncovered except along Alauddin Road. Some encroachments were noted on this drain at the right side of Kashmir Road in Ghouse-e-Azam Colony. It is extensively encroached upon just before the railway line crossing.

Gulberg Drain No. 2

Gulberg Drain No.2 is an earthen trapezoidal drain passing through the Gulberg area. It joins Gulberg Drain No.1 after crossing the Gulberg Main Boulevard near the Home Economics College. Substantial dumping of solid waste was observed in the drain downstream of the Gulberg Main Boulevard. A service road is available on one side of the drain and residential buildings on the other side. Several domestic sewers are directly discharging waste into this drain in addition to the main flow coming from Sunny Flour Mill disposal station, which serves a large part of the Gulberg area.

Gulshan-e-Ravi Drain

This drain serves to carry sewage from Gulshan-e-Ravi pump station to Cantonment Drain. Its total length up to Cantonment Drain junction is 1.0 km. This drain is 4.5 m wide and 1.2 m deep, with a well-maintained brick section in good condition.

In view of the current situation i.e., deterioration of storm water channels and wastage of limited financial resources because of year round pumping into storm water drains, which has also become a permanent public nuisance and has caused

overall degradation of the environment, WASA has decided to connect all the sewerage schemes which are discharging sewage into the storm water channels with new trunk sewers leading to a new disposal/lift station near the existing Gulshan-e-Ravi disposal station. These trunk sewers should be installed at such depths that all the main sewers of the existing sewerage schemes can be connected to the proposed trunk sewers through gravity. It is also important to lay these trunk sewers at considerable depth using trenchless technology in order to avoid interference with, or damage to the existing shallow depth water supply lines and sewers as well as to other buried utility services. This being a pilot project, is expected to be replicated in other sewerage zones of Lahore and other cities of Pakistan as well.

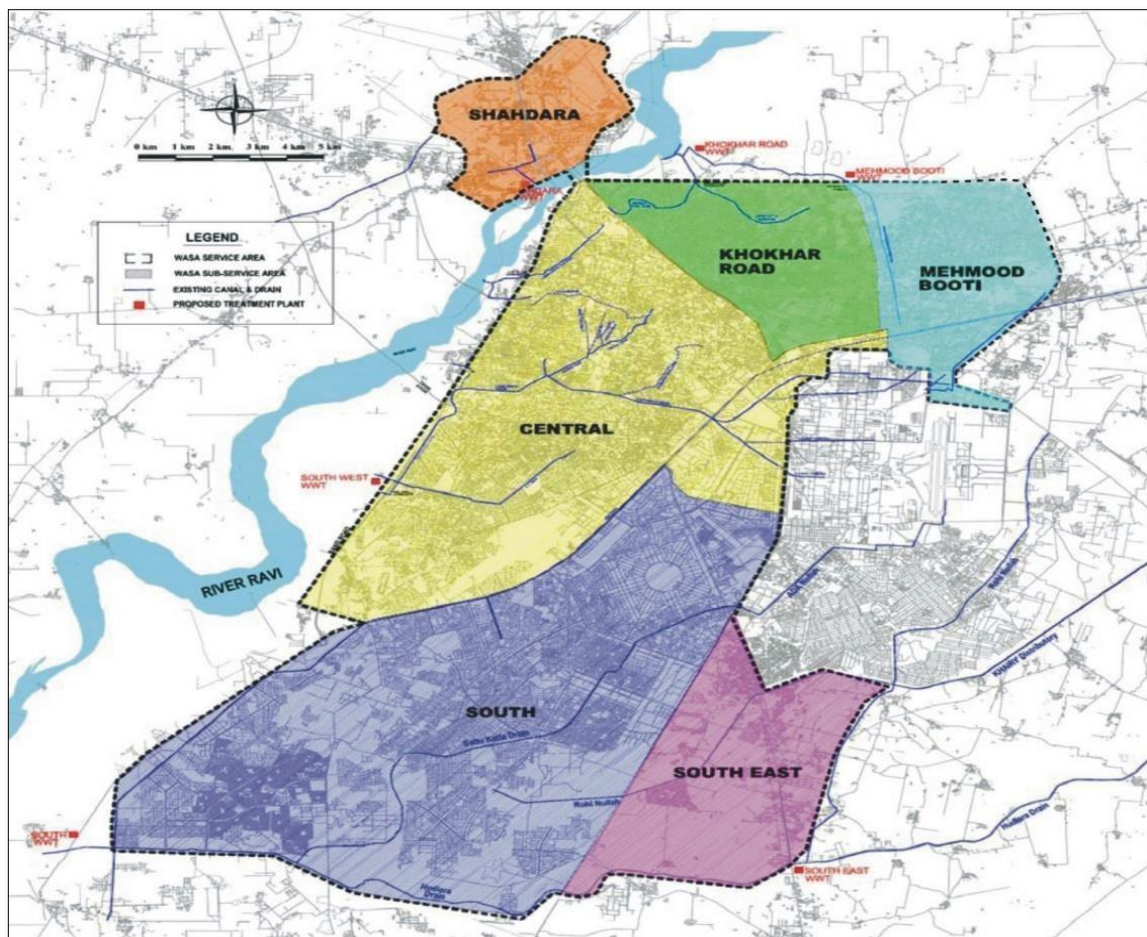


Figure 1: SEWERAGE ZONES OF LAHORE CITY

NEED OF THE PROJECT

Lahore is rapidly urbanizing and transforming into regional urban hub of key economic (commercial, industrial and financial) and socio-economic development and activities. Due to the aforementioned factors, a large influx of migrants seeking greener pastures are migrating to Lahore which is causing the existing systems to become depleted resulting in greater burden on the regulating authorities. The project area is the Central zone, which deficient of a Trunk sewerage system to cater and safely dispose of the sewerage generated. The existing trunk sewers were laid 25-60 years ago which are deficient to cater the flow from the project adjoining areas. The collected sewerage is pumped into the open storm water drains through intermediate lift stations which traverse through the heart of the city causing environmental degradation and odor problems. Moreover, during monsoon season due to inadequate capacity in the storm water drains, the rainfall accumulates in sore points causes flooding in many places and results in degradation of sanitary environment. In lieu of above, WASA, Lahore felt the dire need of a comprehensive sewerage and drainage improvement plan for the area for which consultancy services of M/s NESPAK and M/s MMP were hired for carrying out several studies in past to resolve the aggravating situation of present system of Central Zone.

The studies are listed below:

Master Plan of Central Lahore in 2004 by Consultant NESPAK.

This study involved the construction of Trunk Sewer from Durand Road to Gulshan-e-Ravi Pump Station along with new branch sewers along Davis Road, Empress Road, Montgomery Road and Sanda Road. Besides laying of new sewers as discussed above, sewer cleaning/rehabilitation has been proposed for all trunk sewers within project area to improve the capacity of existing sewerage system.

Study "Laying of Trunk Sewer along Cantonment Drain" in 2009 by Consultant NESPAK

This study involved laying of trunk sewer along the Cantonment Drain. Sewer was proposed in such a way that all the existing disposal stations along Cantonment Drain will be eliminated. Total nine (09) numbers of Disposal Stations were proposed to be eliminated. A new disposal facility has been proposed in the premises of Gulshan-e-Ravi Disposal Station to accommodate the design flow of proposed trunk sewer along Cantonment Drain.

“Master Plan of Sewerage System of District Lahore” in 2019 by Consultant MM Pakistan

WASA, Lahore entrusted the services of Design Consultancy to MM Pakistan (MMP) to carry out the Master Plan study of Lahore City in order to establish the population growth and urbanization trend and ascertain the short- and long-term Investment Plans for future developments for a design horizon upto the year 2040. The Central Zone is the heartbeat of Lahore city and there is a dire need of a Trunk sewerage Network without which the alleviation of the environment would not be possible. The impediment in laying of the Trunk sewer in the Central Zone through conventional approach is not possible due to densely populated area. Therefore, an innovative approach would have to be devised in order to address the problem. These studies were carried out but no practical outcome has been seen so far due to budgetary and other constraints. Due to urbanization and mass development open cut is not feasible anymore. In view of this aggravating situation there is need of project that intends to improve the sewerage system and drainage system of Central Lahore with minimum disruption and disturbance in day-to-day activities. LDA and WASA proposed a project to be executed with trenchless/pipe jacking technology namely “Sewerage System from Larech Colony to Gulshan-E-Ravi, Lahore” and entrusted NESPAK to carry out this project. The proposed project is formulated based on these existing studies with only change in execution strategy of the project.

“Consultancy Services for Design/Review of Design and Resident Supervision of laying of Sewerage System from LARECH Colony to Gulshan-e-Ravi” in 2019 by Consultant NESPAK

Based on the Feasibility Report prepared by the Asian Consultants in 2015, WASA LDA awarded consultancy Services for Design/Review of Design and Resident Supervision of Laying of Sewerage System from LARECH Colony to Gulshan-e-Ravi to NESPAK in 2019. As a first step, NESPAK modified and updated the alignments of the trunk sewers, Branch sewers and Lateral sewers proposed by the Asian Consultants, in view of the infrastructure developments in Lahore after 2015. Similar developments planned for the future were also kept in view while revising the alignments. NESPAK also updated the studies related to micro-tunneling before taking up the detailed design of the project.

The detailed design for the project was prepared by NESPAK on the basis of these modifications and after due deliberations, the consultants submitted the 'Detailed Design Report (Revised) to WASA in July 2019

In their review of the trunk sewer alignment from LARECH Colony to Gulshan-e-Ravi Disposal Station NESPAK proposed certain modification due to some new projects and certain interventions like Metro Bus Project, Orange Train Project, and Express

Highway, which could not have been visualized during the preparation of Feasibility Report by Asian Consultants since conducted prior to NESPAK's study.

NESPAK identified certain conflict points and revised the alignment at the following points:

- Orange Line conflict point at Lakshmi Chowk
- Conflict of Line-A alignment with Orange Line Piling at Multan Road
- Crossing of trunk sewer with Orange Line wat LDA Colony
- Arrangement for conflict with Piling works of Elevated Expressway
- Crossing of Line-A with LRMTS at Qartaba Chowk
- Crossing of trunk sewer below residential area near Garhi Shahu
- Crossing of trunk sewer below encroachments on Chauburji Park Road

PC-I TITLED "SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI LAHORE (THROUGH TRENCHLESS TECHNOLOGY)"

Based on detailed study of Consultant NESPAK, WASA Lahore prepared PC-I based on EPC Mode and as per MRS 1st Biannual, 2019 and the then Market Rates amounting to Rs.14,165.06/- Million. The PC-I was approved by ECNEC tune an amount 14,165.06 Million vide No. SO (UD) 3-4/2019 dated 25.06.2020. (Administrative Approval attached at **Annexure-I**). The PC-I consisted of three packages with Sewerage system of about 28.29 km, brief of which is mentioned hereunder:

Package A (Line-A: Length=15.256 Km)

Package B (Line-B: Length= 13.023 Km)

Package C (Disposal Station: 350 Cfs, 10 Pumps of 50 Cfs each with 3 Standby Pumps).

Instant project depending upon its huge overall cost is going to be financed by Asian Infrastructure Investment Bank (AIIB) through soft loan under Project Name "Lahore Water and Waste Water Management Project".

PROJECT MANAGEMENT CONSULTANCY FOR SCHEME "LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI (THROUGH TRENCHLESS TECHNOLOGY)"

After approval of PC-I, WASA Lahore initiated procurement process of Consultancy services and after completion of same, JV of M/s EBA-ECSP-TEAM-EEC was declared as highest ranked bidder for which contract agreement was made on 25.04.2022 for which formal approval of consultancy cost was granted by Project Steering committee in its 3rd meeting held on 24.01.2022. The Consultancy consists of two phases:

- 1st Phase (Six (06) months- Design Phase)
2nd Phase (Thirty (30) months- Execution Phase).

Project Management Consultancy design phase involves preparation of Preliminary Design, Cost Estimates and Tender Documents. PMC has finalized route alignment survey and preliminary design and Cost Estimates and instant Revised PC-I for the Project. Scope of Consultancy also involves preparation/updating of Environmental and Social Management Plan, Preparation of Tender Documents for hiring of EPC Contractor followed by its work Contract and supervision of work after preparation of detailed design by contractor. Formulation of Revised PC-I was necessitated owing to upward revision in cost of Project Management Consultancy after completion of Procurement process and as per decision taken in 3rd Project Steering Committee meeting held on 24.01.2022 and Change in Alignment of original route after detailed working by PMC based on prevailing site conditions, updated MRS rates and to account for all flows disposing off into Cantonment drain.

After thorough on-site surveys and field investigations, catchment areas essential for augmentation of sewerage system of central Lahore were delineated. Accordingly, on the basis of updated design and decided route, revised PC-I amounting to PKR 47,896.499 Million was submitted to HUD & PHED on 09.06.2023.

Later on, PC-I of WASA trunk sewer project was also returned back on 11.08.2023 by P&D Board for making requisite amendments due to change in alignment-cum-design and updating the cost estimates on new Market Rate System (MRS).

Issues regarding change in alignment were raised owing to fact that LDA has also formulated a project of "Elevated Expressway (EEW) from Main Boulevard Gulberg to Motorway (M2)" for which Pre-PDWP meeting was held on 07.06.2023 in P&D board. In the meeting, matter regarding conflict points of LDA project with WASA project of Trunk Sewer also came under discussion. It was discussed in meeting that Line B & Line C of the micro-tunneling project have major overlapping with the EEW project starting from Main Boulevard Gulberg to Babu Sabu along the Cantonment Drain. A follow-up meeting in this regard was also held at 8-Club Road, GOR-I under the chairmanship of Chief Minister Punjab on 21.06.2023 which was attended by Chief Secretary Punjab, Chairman P&D Board, Secretary HUD & PHED, Commissioner Lahore, MD WASA and Chief Engineer LDA. PMC informed that owing to interference with the EEW, original alignment of trunk sewer would not remain feasible due to the reason that limited vertical clearance under EEW (5.1 m) would not allow crane to operate to construct multiple access shafts, lower MTB machine and lay 3500 mm large diameter pipes.

Moreover, it was elucidated that in case Project of Elevated Express Way (EEW) is executed prior to caption project of WASA, difficulties regarding O&M of trunk sewer may arise in future owing to insufficient vertical clearance. The PMC also apprised the forum about the need of a feasibility study through the realigned route of Line B &

Line C supported with new social and topographic surveys, additional geotechnical boring, and hydraulic modelling. Matter regarding interference of projects of Punjab Mass Transit Authority (PMA) like Blue and Purple Line projects of mass transit along Allama Iqbal Road and Bahawalpur Road were also discussed. PMA apprised the forum that cabinet has approved that no development project shall be planned in the alignment of PMA Metro Line projects

In addition to above, AIIB during its mission of Lahore dated 17th-18th August 2023 also sought updated progress of instant project, wherein, PMU and PMC apprised the forum regarding alternate viable options of laying of trunk sewer by avoiding conflicts points owing to construction of EEW. As per PMC, alternate routes for Line-B & Line-C would need to be studied in detail in order to opt most feasible route.

Hence, in order to conduct feasibility study for alternate routes and to update documents i.e. Preliminary Engineering Design Report, Environmental and Social Impact Studies (ESIA, RP & LRP), and Tender Documents, services of consultant were need to be hired for which Budget Estimate and TORs were prepared and presented before 5th Project Steering Committee meeting held on 22.01.2024. In the meeting, it was decided to carry out Feasibility Study of alternate route of the instant project.

Accordingly, PMC was engaged on 30.04.2024 to carry out Feasibility of the Realigned route after approval of TORs, Cost Estimates and Budget from AIIB. As per this Amendment-II, PMC shall carryout feasibility study for alternate routes and will update documents i.e. Preliminary Engineering Design Report, Environmental and Social Impact Studies (ESIA, RP & LRP), and Tender Documents.

JUSTIFICATION OF PROJECT:

Instant PC-I is second Revised PC-I prepared against ADP Scheme falling at GS No 3036 Fy 2024-25 titled "Sewerage System from LARECH Colony To Gulshan-e-Ravi Lahore (through Trenchless Technology)" which was approved by ECNEC amounting to Rs.14,165.06 Million in year 2020.

Following are the reasons which necessitated revision in cost of instant scheme after updated working by Project Management Consultant (PMC) (JV of American Firm M/s EBA with EEC Global, ECSP & TEAM):

1) Increased Designed Flow owing to Change in Alignment, increased catchment area & Beneficiary Population

Main objective of the instant scheme is to reinstate Cantonment Drain as storm water carrier by eliminating all lift stations that are outfalling into the drain thus by restoring recreational activities at Cantonment Drain and natural ecology. Previously NESPAK proposed elimination of 12 stations that were outfalling into cantonment drain in contrary to which latest PC-I involves elimination of 22 lift stations of WASA service area including consideration of flow of 12 lift stations of Lahore Cantonment Board

cum Walton Cantonment Board, thus almost all of the flow outfalling into cantonment drain through disposal stations has been considered. About 688 cusec design flow for planning horizon of year 2050 and Targeted population of about 2.35 Million (15658.02 acres) has been taken into consideration while preparing instant PC-I in contrary to 350 cusec designed flow (Planning horizon 2027) with serving population of about 0.8 Million (approx. 6000 acres) in previously approved PC-I thus a bigger disposal station (848 cfs ultimate capacity) will be constructed as compared to (550 cfs ultimate capacity) disposal station proposed by NESPAK earlier in previously approved PC-I.

2) Increased Trunk Sewer Depth & Diameter

Three Trunk lines (Line-A, Line-B & Line-C) will be laid under instant scheme varying in diameters from 1500 to 2400mm and finally 3500mm terminating at proposed Gulshan-e-Ravi Disposal of about 848 cfs ultimate capacity as compared to sizes of 1500mm, 1800mm and 2400 mm utilized by NESPAK in previous PC-I. Resultantly, depths of sewer also increased from 10.63 m (Karachi Pathak) to about 17.28 m (New Gulshan-e-Ravi DS) for instant scheme as compared to depths of 11.5m (LARECH colony) to 14.36m (New Gulshan-e-Ravi DS) in previous. Hence, increased in size and depth of trunk sewer to cater more catchment areas eventually contributed towards increased in overall cost. Further construction methodology from caisson to sheet piles for safe and expeditious working and sizes of Jacking and Recipient shafts for different pipe sizes of 1050mm, 1500mm, 2400mm and 3500mm have also been revised under instant PC-I.

3) Increase in Quantity of Micro Tunnel Boring Machines (MTBMs Quantities)

Three MTBMs were utilized earlier by NESPAK of sizes 1500mm, 1800mm and 2400mm however, keeping in view the updated working and after formulation of preliminary design by PMC, it has been envisaged that at least Six MTBMs (02 of 2400mm, 01 of 1500mm and 01 of 3500mm and 02 of 1050mm) will be required to complete the project within stipulated period upto June 2027.

4) Change in Market Rates system from 1st Biannual 2019 to 2nd Biannual 2024 & Huge variation in Costs of Non-MRS Items

Previous PC-I was approved for an amount of Rs 14,165.06 Million in year 2020, wherein MRS of 1st Biannual 2019 was used for rates of standardized items. Now instant PC-I has been framed on MRS of 2nd Biannual 2024 resultantly causing increase in overall amount. Similarly, after soaring inflation from 2019 to 2024 (5 years), rates of Non MRS items have also been increased multiple times, which is also major reason behind higher costs of instant PC-I.

5) Inclusion of Items like intake structure, Drop Structure, Ventilating Shafts and Dedicated Grid Station for Proposed disposal station

In earlier PC-I, several important components like intake structure (containing inlet chamber, screening chamber, outlet chamber, proportional weir), drop structures were not taken into consideration while making connection of existing eliminated lift stations, however keeping in view their pivotal importance same including their costs has been taken into consideration for (22+12) eliminated lift stations. Similarly due to higher overall electric load of 10 MW owing to proposed construction of new 848 cfs disposal station including allied accessories at Gulshan-e-Ravi and to meet existing electric load of existing as per LECSO policy; an independent Grid station will be constructed.

6) Revision in Project Management Consultancy Cost

As per previous approved PC-I, cost of Project Management Consultancy of 354.72 Million was approved, however, after completion of Procurement process of project management consultant and as per decision taken in 3rd Project Steering Committee meeting held on 24.01.2022 in P&D Department, revised consultancy cost of 707.43 Million (exclusive of taxes) was approved. Similarly in order to ensure sufficiency of Geological data for facilitating the cutting head design of MTBM and to curtail associated risks that might be taken by EPC in its bid, an Amendment No.1 was awarded to PMC with Cost of 81,834 MUSD+ 42,384,477 MPKR after approval from AIIB. Similarly, in order to avoid conflict with Elevated Express Way Project of LDA, Amendment No.2 was awarded to PMC on 30.04.2024 after approval from PSC in its 5th Meeting dated 22.01.2024 and after approval of Scope of Work and Budget from AIIB for contract amount of 283,960 USD+ 47,238,382 PKR. Thus owing to aforementioned factors and after inclusion of Price inflation index as per contract agreement for Phase-II overall consultancy cost was increased.

All these factors/justifications as highlighted in foregoing paras resulted in overall increase in project Cost, component wise cost difference summary of same is mentioned alongwith Cost Abstract (Refer Page No 82 of instant PC-I)

BRIEF COMPARISION OF NESPAK (2019) Vs PMC PROPOSED PC-I

Sr No	Description	NESPAK (Approved PC-1) (2019)	PMC (Proposed PC-1) (2024)	Difference	
				Values	%age
1	Planning Horizon	2045	2050	5	0.25%
2	Catchment Area (Acres)	8,166	15,658	7492	92%
3	Served Population	1,221,613	2,355,442	1,133,829	92%
4	Design Flow (Cusec)	350	688	338	97%

5	<u>Length of MTB (Mtr)</u>				
	1500 mm	11,391	2,087	-6,583	-82%
	1800 mm	2,759	0	-2,759	-100%
	2400 mm	14,059	11,582	-2,477	-18%
	3500 mm	-	5,337	5337	100%
	Total Length Main Trunk Sewer through MTBM in meters	28,209	19,006	-4,374	-33%
6	Length of Laterals to be laid through MTBM in meters (1500mm & 1050mm)		13,204	13,204	100%
	Length of Laterals to be laid through open cut in meters (450mm)	805	159	-543	
	Total Length of Route in meters (through MTBM + Open Cut)	29,014	32,369	3,458	12%
7	Jacking Shafts (Nos)	51	82	31	61%
8	Receiving Shafts (Nos)	50	83	33	66 %
9	No. of Disposal Stations	12	22 (Direct) (with provision in flow of 12 DS)	10	83%

Significance of Tunnel Boring Vs Open Cutting

The significance of Tunnel boring Vs Open Cutting is illustrated hereunder:

Table 1: Comparative Analysis of Open Cut & Trenchless Methods

Sr. No	Open Cut Method	Trenchless Technology
1.	Open cut method often involves risk Pipe jacking are safer methods of working	Pipe jacking are safer methods of working
2.	Dismantling of road takes place	Dismantling of road only requires for launch and reception shafts for entrance of pipe jacking machine. Substantial cost saving.
3.	Slow Construction, often takes more time	Speedy Construction
4.	Public Inconvenience	No inconvenience to public.
5.	More Traffic disturbance	Traffic disruption are reduced or eliminated, Mobility above ground is not affected
6.	More Environmental effects, soil disturbance	Minimum environmental effects less soil is Disturbed /excavated and movement.
7.	Shifting of all underground utilities	Minimum shifting underground utilities, services Cost effective
8.	More Transportation & Rehandling of Materials	Reduce the quantities of incoming and outgoing materials
9.	Maximum visual impact to the local Population	No impact to the local population
10.	Less reliability More cost	High degree of reliability and cost Effectiveness.

UPDATED SCOPE OF WORK

GENERAL

WASA alongwith its PMC determine individual catchment areas of all the existing disposal stations which pump sewage effluent into the Cantonment Drain and its branches. For this purpose, extensive site visits were made and detailed discussions were held with Operation and Maintenance staff of WASA, in order to establish the areas contributing to each disposal station. WASA also required some additional schemes and areas to be included in the design of the trunk sewers. Accordingly,

significant increase in the total catchment area has been observed as compared to the previously proposed scheme. The catchment area of each disposal station has been marked on the project area map to estimate the sewage flows as discussed in the following.

Contributing Areas of Line-A

In past, Line-A was proposed to start from the LARECH Colony Disposal Station (DS), however, in order to accomplish the desired objectives of this project i.e. separation of sewage from cantonment drain, it was decided during various in-house meetings that the proposed project should cater for the sewage effluent from the Harbanspura scheme as well, which is presently under execution. The proposed Line-A will therefore start from Karachi Phatak on Allama Iqbal road, where the outfall sewer of Harbanspura scheme will deliver its sewage discharge to the proposed trunk sewer Line-A. Besides, discharges from Twelve (12) disposal stations under the control of Cantonment Board, which are presently pumping sewage into the Cantonment Drain, have also been included in the project.

Accordingly, Line-A will now start from Karachi Phatak, where it will be connected with the 72 inch dia outfall sewer of the Harbanspura Sewerage Scheme, which is to be laid by WASA through ADP scheme. This 72" dia outfall sewer will bring sewage effluent from Harbanspura area, Taj Bagh housing scheme, Manzoor Colony, Taj Pura area, Sally Town, Jalal Colony, Haji Pura area, Aziz Bhatti Town, Jorray Pul area and Al-Faisal Town.

Besides, under this project, provision has been made in Line-A for including the sewage discharge from the disposal stations under the jurisdiction of Walton and Lahore Cantonment Board, as named in Table 2, A lateral from the Infantry Road DS will also connect this trunk sewer before canal road crossing. This combined sewer will also account for flow from Bird market Disposal of WASA

On the other side, a branch sewer will start from Larech Colony DS and will collect sewage from Ghazi Mohallah DS through a small lateral. At the junction of Allama Iqbal Road and Durand Road a lateral from Muhammad Nagar D/S will discharge its sewage effluent into the Trunk Sewer Line-A.

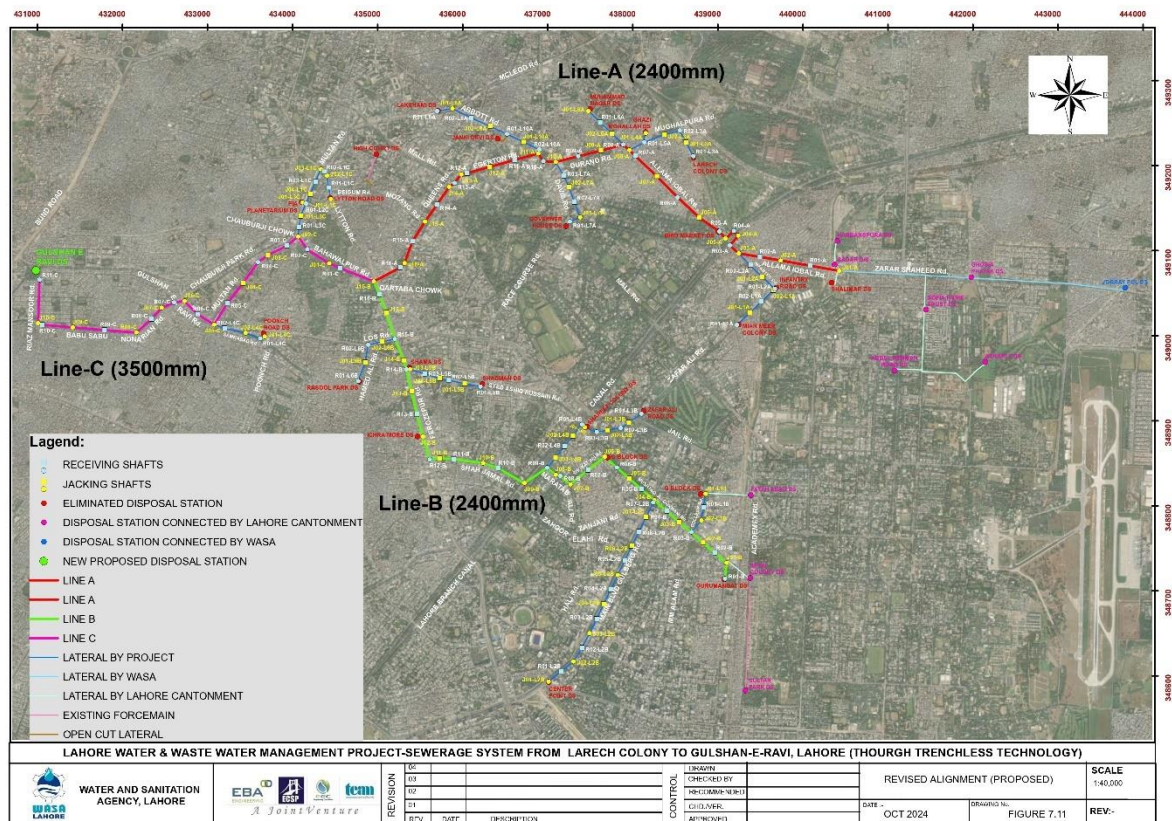


Figure 2: Layout plan of Trunk Sewers of LARECH Project

At Durand/Davis Road junction, a lateral from Governor House, will join Line-A. Another lateral from Lakshmi Chowk will join Line-A on Edgerton Road. Further down, a lateral from Mozang Adda, will be laid and it will join Line-A at the junction of Mozang Road and Queens Road

Contributing Areas of Line B

Lateral sewer from Gurumangat Road DS and another lateral from Q-Block Gulberg DS will join at the starting point of Line B. On the way sewage discharge from Gulberg G- Block DS will be collected. A lateral sewer from the Cantonment Board's Mian Mir DS, will also be connected to Line-B that will also pick up discharge from the Zafar Ali Road DS after which, it will cross canal road. Discharge from Shareef Colony DS, will be added to Line-B, before it crosses the canal. Further down, the sewage effluent from Shadman DS will also be collected by Line-B through lateral.

In addition, a provision has been made in the Line-B's capacity for 25% of the sewage discharge of PIC, Mental Hospital and GOR III, and area on the right side of Jail Road,. A provision of 25% of the sewage discharge from the area on the right side of Jail Road measuring in the capacity of Line-B for allowance of High-Rise Commercial Buildings. Moving further ahead, sewage from Shama D/S, will be collected by Line-B beside discharge from Ichra D/S including flow from another lateral from Rasool Park D/S.

Contributing Areas of Line-C

Line-C will start from Qartaba Chowk, where Line-A and Line-B combine. A lateral from Lytton Road D/S, will be laid. It will join Line-C at Chauburji Chowk. Another lateral from PIA Planetarium DS will also join Line-C at Chauburji Chowk.

One of the major branch sewer under the project, starting from Poonch Road DS will join the Trunk Sewer Line-C on Main Boulevard. A provision for 25% of additional sewage effluent from GOR II and Chauburji Quarters has also been made in the capacity of this branch sewer. Line-C will be laid along Main Boulevard Gulshan-e-Ravi upto Gulshan-e-Ravi Disposal station by passing along Babu Sabu Nonarian Road and Riaz Mansoor Road.

1.10 TOTAL CATCHMENT AREA OF LARECHS PROJECT

The total catchment area of this project is 15,658.02 acres (6,339.28 ha) and about 22 Disposal stations of WASA will be eliminated after implementation of instant project. In addition, flow of 12 Disposal stations of Lahore cantonment Board and Walton Cantonment Boards have also been considered, same are mentioned hereunder:

Table 2: Disposal Stations proposed to be eliminated under LARECHS Project (22+12 No)

Sr No	Disposal Station/Area	Sr No	Disposal Station/Area
Line- A		10	Rasool Park DS
1	Mian Mir DS	Line- C	
2	Infantry Road DS	1	Lytton Road DS
3	Bird Market DS	2	PIA Planetarium DS
4	LARECH Colony DS	3	Poonch Road DS
5	Ghazi Muhallah DS	LCB+WCB (Flow Considered)	
6	Muhammad Nagar DS	1	Joray Pull DS
7	Governor House DS	2	Ghosia Phattak DS
8	Lakhshmi Chowk DS	3	Askari 9 DS
9	Janki Devi DS	4	Abdur Rehman Road DS
Line- B		5	Sofia Trust DS
1	Gurumangat DS	6	Harbanspura DS
2	Q – Block DS	7	Sadar DS
3	Center Point DS	8	Shalimar DS
4	G Block DS	9	Sultan Park DS
5	Zafar Ali Road DS	10	Khan Colony DS
6	Shareef Colony DS	11	Fateh Abad DS
7	Ichra DS	12	Askari Villas
8	Shadman DS		
9	Shama DS		

ALIGNMENTS OF TRUNK SEWERS AND LATERALS

Currently, most the sewage effluent from Lahore Cantonment and WASA service area of the Central Zone is being disposed of without any treatment into Cantonment Drain and its branches, through several disposal points. Besides, at numerous places, sewage effluent from areas close to the existing drains is being directly discharged into the nearby drains. There are twenty-two (22) disposal stations under the jurisdiction of WASA and Twelve (12) disposal stations under the jurisdiction of Cantonment Board which are presently pumping the sewage directly into the Cantonment drain and its branches.

The proposed project requires a trunk sewer system to be designed in such a way that all the existing disposal stations along Cantonment Drain and its branches should be eliminated and sewage is diverted to the proposed trunk sewers for improving the health and environmental conditions in the Central Zone of Lahore.

Two main trunk sewers i.e., Line-A and Line-B are proposed for collecting sewage effluent from northern and southern portions of the Central Zone of Lahore. The two trunk sewers will terminate at Qartaba Chowk and from there onwards up to the proposed Ultimate Disposal Station location at Gulshan-e-Ravi, a larger size Line-C will be provided to carry the combined discharge of Line-A and Line-B. Pumped discharge from the Ultimate Disposal Station will be led to an outfall channel under this project, which will finally discharge it into the proposed collector channel (which is to be executed under another project of WASA) for onward transmission to the proposed Sewage Treatment Plant at Babu Sabu.

PHYSICAL PHASING OF PROJECT

Total Proposed Sewer Length= 32.369 km

Total Pumps to be installed 16 No (53 cfs each)

Sr.	Description of Packages	Year Wise Physical Activities		
		2024-25	2025-26	2026-27
1	Laying of Sewer Line from Karachi Phatak and Gurumangat Gulberg to Gulshan-e-Ravi (through trenchless technology) and connections with existing disposal stations to be eliminated	20% (6.47 km)	40% (12.95 km)	40% (12.95 km)
2	Construction of Ultimate Disposal Station of 688 cfs (designed Capacity) at Gulshan-e-Ravi and rehabilitation of outfall channel along Allied Works.	20% (Const. of Dry well)	40% (8 No pumps)	40% (8 No Pumps)

FINANCIAL PHASING OF PROJECT

(Project Cost: PKR 49,270,400,000/-)

Sr.	Description	Year Wise Financial Utilization (%)		
		2024-25	2025-26	2026-27
1	Laying of Sewer Line from Karachi Phatak and Gurumangat Gulberg to Gulshan-e-Ravi (through trenchless technology) and connections with existing disposal stations to be eliminated	9,854.08 M	19,708.16 M	19,708.16 M
2	Construction of Ultimate Disposal Station of 688 cfs (designed capacity) at gulshan-e-ravi and rehabilitation of outfall channel along allied works			

2nd REVISION OF PC-I TITLED "SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI LAHORE (THROUGH TRENCHLESS TECHNOLOGY)"

2nd Revised PC-I of instant scheme was placed before PDWP in its 26th Meeting held on 09.09.2024 and after extensive discussion same was cleared at cost of PKR 49,270.40 Million for onward transmission to CDWP/ECNEC for approval (Minutes of Meeting attached at **Annexure-2**)

6.2 SECTORAL SPECIFIC INFORMATION:

Sectoral Specific Information

The project is in line with Sustainable Development Goals- 6.1, National Water Policy, 2018 and Punjab Sector Development plan 2014-2024 for Drinking Water, Sanitation and Hygiene.

1. Sustainable Development Goal 6 (SDG 6 or Global Goal 6)

SDG is about "clean water and sanitation for all". It is one of 17 Sustainable Development Goals established by the UN General Assembly in 2015, the official wording is: "Ensure availability and sustainable management of water and sanitation for all. The goal has eight targets to be achieved by 2030. Safe and affordable drinking water

- i. End open defecation and provide access to sanitation and hygiene
- ii. Improve water quality, wastewater treatment, and safe reuse
- iii. Increase water-use efficiency and ensure freshwater supplies
- iv. Implement IWRM (Integrated Water Resources Management)
- v. Protect and restore water-related ecosystems
- vi. Expand water and sanitation support to developing countries
- vii. Support local engagement in water and sanitation management

2. National Water Policy, 2018

Plans and initiatives shall be undertaken to progressively provide access to clean and safe drinking water and sanitation facilities to the urban and rural population of the country. The Rural Water Supply and Sanitation Services shall be priced at affordable rates. Drinking water and sanitation plans will be adopted in line with the National Drinking Water and Sanitation Policies and Sustainable Development Goals.

3. Punjab Sector Development plan 2014-2024 for Drinking Water, Sanitation and Hygiene

The Punjab Water Sanitation and Hygiene Sector Strategy provides a strategic direction for integrated response in the sub sectors of drinking water, sanitation and solid waste management. It articulates the long term aspirations of the Government of the Punjab as it forges ahead with determination to ensure that its populace has universal access to improved water and sanitation.

4. Punjab Growth Strategy 2023

As per Punjab Growth Strategy 2023 section 2.2 " Targets for the Punjab Growth Strategy 2024," it has been planned to Extend sewerage system lines to provide 70 percent

coverage in cities and towns and 40 percent coverage in rural area”, which is directly correlated with the objectives of this project which has been formulated to provide effective sanitation facilities to about 2.35 Million People of Central Zone of Lahore City

7. CAPITAL COST ESTIMATES:

Financial Components: Capital
Cost Center:OTHERS- (OTHERS)
Fund Center (Controlling):LE4280

Grant Number:Engineering - (PC220036)
LO NO:N/A
A/C To be Credited:Assignment

PKR Million

Sr #	Object Code	2024-2025		2025-2026		2026-2027	
		Local	Foreign	Local	Foreign	Local	Foreign
1	A05270-To Others	4,826.410	0.000	0.000	0.000	0.000	0.000
Total		4,826.410	0.000	0.000	0.000	0.000	0.000

8. ANNUAL OPERATING COST (POST COMPLETION)

Financial Components: Capital
Cost Center:OTHERS- (OTHERS)
Fund Center (Controlling):LE4280

Grant Number:Engineering - (PC220036)
LO NO:N/A
A/C To be Credited:Account-I

PKR Million

Sr #	Object Code	2027-2028		2028-2029		2029-2030		2030-2031		2031-2032	
		Local	Foreign	Local	Foreign	Local	Foreign	Local	Foreign	Local	Foreign
1	A05270-To Others	487.409	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Total		487.409	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

9. Demand and Supply Analysis:

Instant Scheme is based on need based assessment of about 2.35 Million inhabitants of Lahore City living in Central and South West areas of Lahore. Demand Assessment for Instant scheme has been envisaged based on per capita water consumption. The Project is expected to have designed capacity of 688 Cusec with 848 Cfs ultimate disposal capacity.

10. FINANCIAL PLAN AND MODE OF FINANCING

10.1 FINANCIAL PLAN EQUITY INFORMATION:

N/A

10.2 FINANCIAL PLAN DEBT INFORMATION:

Instant project of "Laying of Sewerage System from LARECH Colony to Gulshan-e-Ravi (through trenchless technology)" amounting to Rs. Rs.49,270.40 MILLION is 2nd revision of already approved project from ECNEC of which administrative approval of PKR 14165.06 Million, issued vide No SO(UD)3-4/2019 dated 25.06.2020. The Gestation period of revised project is 36 Months and same falls under umbrella of AIIB funded "Lahore Water and Waste Water Management Project". AIIB's Investment Committee has approved the appraisal of subject project on 19 Oct, 2022 authorizing the team to negotiate a loan of USD 329 M with the Government of Pakistan. The loan shall be paid in accordance with amortization schedule decided upon loan negotiations.

10.3 FINANCIAL PLAN GRANT INFORMATION:

N/A

10.4 WEIGHT COST OF CAPITAL INFORMATION:

AIIB's Investment Committee has approved the appraisal of subject project on 19 Oct, 2022 authorizing the team to negotiate a loan of USD 329 M with the Government of Pakistan. Weight Cost of instant Project is Rs. Rs.49,270.40 Million

11. PROJECT BENEFITS AND ANALYSIS

11.1 PROJECT BENEFIT ANALYSIS INFORMATION:

Social Benefits of Project

BRIEF ON SOCIAL BENEFITS

Instant Project has multiple tangible and intangible social benefits and same has been envisaged with moto to create positive impact on environment and restoration of natural ecology and recreational activities in the areas located in suburbs of primary drains like Cantonment Drain as well as Secondary drains like Central Drain, Gulberg Drain, Lakshmi Drain etc., After implementation of project and diversion of sewage flow from independent house hold lateral sewers (that are directly disposing sewage waste into Cantonment drain) and catering of flow from forcemains of lift stations to Trunk sewer to be laid under this project upto Gulshan-e-Ravi Disposal station, livelihood activities in the surrounding areas will be restored. Further, issues concerning to under capacity and old sewerage system (some laid in pre-partition period) will also be redressed. The improved and augmented sewerage system will be sufficient to accommodate larger quantity of sewage flows by encompassing all those chronic areas that were deprived earlier as far as better sanitation facilities are concerned.

On the other hand, project will also assist in mitigation of storm runoff, averting the traffic disruption issues and safe disposal of collected water especially during heavy rainy days. Saving in Energy Costs, Manpower Requirements after elimination of intermediate pumping stations i/c Generators and Extra Costs that were incurred on R&M of Drains cum Sewers earlier will also be curtailed. Chances relating to Road damages will also get meager after implementation of instant project

Similarly in addition to above WASA Lahore has planned to construction a wastewater treatment plant of about 178 MGD (2035) and 225 MGD (2045) at Babu Sabu which profoundly will accommodate flows originated from Central Zone of Lahore city., hence, after reinstating the cantonment drain into its initial condition as storm water carrier, it will be simplifier to divert sewage outflows from Gulshan-Ravi Disposal to proposed treatment plant, thus avoiding further cost implications for separate conveyance system of this treatment plant without compromising on larger design capacity issues.

11.2 ENVIROMENTAL IMPACT ANALYSIS:

ENVIRONMENTAL IMPACT ANALYSIS

It is pertinent to mention here that as per AIIB Environmental and Social Framework, Environmental and Social Appraisal of the Projects are mandatory to be carried out, similarly Environmental and Social Policy (ESP) and Environment and Social Standards (ESSs) also shed light on same obligatory requirements before implementation of large-scale projects. The objective of this overarching policy/framework is to facilitate achievement of these development outcomes, through a system that integrates sound environmental and social management into Projects.

It further envisages the following:

- Support decision-making by the Bank.
- Provide a robust structure for managing operational and reputational risks of the Bank and its shareholders in relation to environmental and social risks and impacts in Projects.
- Provide for environmental and social screening and categorization of Projects.
- Analyze potential environmental and social risks and impacts of Projects.
- Identify actions to avoid, minimize, mitigate, offset or compensate for environmental and social impacts of Projects.
- Support integration of environmental and social management measures into Projects.
- Specify environmental and social management provisions to be included in agreements governing Projects.
- Provide a mechanism for public consultation and disclosure of information on environmental and social risks and impacts of Projects.
- Provide for monitoring and supervision of environmental and social management measures under Projects.
- Facilitate development and dissemination of lessons learned from Projects to improve environmental and social management practices.

The Punjab Environmental Protection Act categorizes the proposed projects as Schedule-II (F). Whereas, in accordance with the AIIB's Environmental and Social Policy (ESP), the proposed Potential Projects project has been assigned Category 'A'

A project is categorized “A” if it is likely to have significant adverse environmental and social impacts that are irreversible, cumulative, diverse or unprecedented. These impacts may affect an area larger than the sites or facilities subject to physical works and may be temporary or permanent in nature

As per Environmental and Social Framework Policy of AIIB, Following Mitigation Plan have been formulated for

**SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN-E-RAVI, LAHORE/
TRUNK SEWER COMPONENT:**

- **ESIA** (Environmental and Social Impact Analysis study) for complete Project of LARECHS Colony.
- **LRP** (Livelihood Restoration Plan) for contributory Catchment Areas of Trunk Sewer whom business/livelihood activities would be disturbed during project execution.
- **RP** (Resettlement Plan) for Construction Of Disposal Station at Gulshan-e-Ravi in Lahore City
- **GAP** (Gender Action Plan) for complete Project of LARECHS Colony.

EIA of Previously approved PC-I of "Trunk Sewer from LARECH Colony to Gulshan-e-Ravi Disposal station has already been approved and NOC granted by Environmental Protection Department Government of the Punjab **Annexure 03**.

Environmental and Social Impact study as well as Livelihood Restoration plan for updated cost and revised route alignment as reflected in instant 2nd Revised PC-I have already been disclosed on WASA Website after approval by AIIB Social and Environmental safeguard specialists. Further all social costs as being reflected in aforesaid social studies were also presented in 4th Project Steering committee meeting held on 28.09.2022 at P&D Department and were principally approved.

Summaries of ESIA study, LRP, RP and GAP studies are appended in **Annexure 04, 05 06 and 07** of this PC-I respectively. It is worth mentioning here that these environment

and social documents have been prepared on the Project Feasibility Report of Revised Alignment. The impacts and costing assessed in these documents are as per earlier prepared feasibility design requirement. Since these documents are live and will be updated in case of any minor or major change in design. The costing and impacts of the project will also be revised / updated as per the new design requirement.

Cost Summary is provided in below table:

Sr. No.	Description	Amount	Amount
		(PKR)	(PKR)
			(in Million)
1	ESIA	29,284,320.00	29.28
2	RP	24,409,665.00	24.41
3	LRP	68,143,152.00	68.14
4	GAP	8,615,000.00	8.62
5	E&S consultant for Phase-2 (E&S expert & support staff and GAP expert) (Annex-8)	60,555,020.00	60.56
Grand Total		191,007,157.00	191.01

Air Quality and Climate Change Aspects

Climate change is considered to be one of the main challenges to urban wastewater systems. The operation of sewerage scheme and disposal station will result in generation of gases including H₂S and CH₄ (due to anaerobic conditions). H₂S is a foul-smelling poisonous gas which might be harmful for the sewer-men and could be fatal sometimes. CH₄ is natural gas also referred to as a greenhouse gas is one of the main greenhouse gas contributors to global warming with a lifespan of about 12 years and a global warming potential of roughly 21– 23 times higher than carbon dioxide. Pakistan is signatory to the United Nations Framework Convention on Climate Change (UNFCCC) and adopted Kyoto protocol in 1997. Under the Kyoto Protocol, Pakistan is committed to reduce GHG

emissions into the environment to prevent interference with climate change. The GHG emissions for the proposed project are a concern due to the anaerobic conditions in the sewer lines and wet wells of the disposal station; whereas, the gas emissions during the conveyance of wastewater will be eliminated by the trench sewer. There is always a probability of malfunctioning of the system and uncontrolled leakage of these gases.

The existing sewerage system is old and outdated. It will be replaced with underground wastewater conveyance system through MTBM. This would impede the nuisance of environmental problems associated with open drains as well as the new system will consume 29% less electricity. The electric load of all the proposed 22+11 eliminated and to be eliminated pumping stations was taken and compared with the total electricity consumption of the proposed solution. An analysis between 2025 to 2050 was drawn (Annexure II.) and a net reduction of 127,612 Tons CO₂ eq in GWP was observed, yielding carbon credits of 127,612 unit.

Comparison of Power Consumption & CO ₂ Emissions b/w Existing and Proposed Sewerage System FOR 25 YEARS (2025-2050)					
Descriptions	Unit	Existing System	Proposed System	Difference	%age Difference
Power Consumption	KWh	426,762,243	329,468,783	97,293,460	23%
Fuel Consumption From Gensets	Gal	3,116,094	0	3,116,094	100%
CO ₂ Emissions Total	Kg	394,469,748	280,048,466	114,421,283	29%
Power Generation (450 KW PV)	KWh	0	15,519,592	15,519,592	100%
CO ₂ Saving (450 KW PV)	Kg	0	13,191,654	13,191,654	100%
*Tons of CO ₂ / Carbon Credit	Tons/Credits	394,469.75	266,856.81	127,612.936	32%

During operation phase, operation of stand-by generator and un-necessary idling of diesel run will also result in the emission of pollutants (CO, Smoke, NO₂, SO₂ CO₂ and PM₁₀) into the air, thus deteriorating air quality. These emissions may adversely affect the health of the nearby residents and working staff.

This impact can be categorized as negative, local, medium, long-term and definite.

Following mitigation measures will be adopted:

- To mitigate the pollutants emissions use of low sulfur fuel will be ensured;
- Strict prohibition on re-used fuel;
- Location of generators will be carefully selected;
- Plantation of trees along the disposal and pumping stations to create a buffer zone that will help in absorbing the emissions;
- Use of gas generators (if possible) will be preferred for low emissions;
- Use of solar panels (renewable energy source) for running new installed pumps and generators, as it will save the energy;
- The height of the generator stacks will be high enough to disperse the emissions in the air;
- Provision of budget for regular monitoring of ambient air quality in accordance with PEQS, 2016;
- Improved monitoring, planning, and maintenance of the sewerage system by WASA;
- Sewer-men will cover their faces with mask while entering the sewer for cleaning or maintenance purposes;
- Use gas detector before inspection;
- If possible, use different mechanisms to capture CH₄ and H₂S that can be used commercially; and Plan and execute appropriate tree plantation along the disposal station.

11.3 ECONOMIC ANALYSIS:

Economic and Financial Analysis elucidated hereunder

11.4 FINANCIAL ANALYSIS:

The Projected stream of economic benefits over the economic life of the project has been compared to the estimated stream of economic costs by bringing the two streams to a uniform basis through the process of discounting. To judge the economic viability of the project, the following economic indicators have been computed:

1. Net Present Value (NPV).
2. Benefit Cost Ratios (B/C Ratio).
3. Economic Internal Rate of Return (EIRR)

Results of economic cum financial analysis for each component are illustrated in forthcoming section

ECONOMIC ANALYSIS

SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCH-LESS TECHNOLOGY)

1. GENERAL

The city of Lahore is population-wise the second largest city of Pakistan after Karachi. It is undergoing rapid modernizing process, requiring substantial improvement in infrastructure for facilitating and improving the living conditions of its residents. Among others, water supply, sewerage and drainage are some of the most important utilities for sustaining the public life in general and public health in particular. For meeting this requirement, Water and Sanitation Agency (WASA) was established by the Lahore Development Authority (LDA) in 1976, Government of Punjab. Lahore WASA has launched the “Lahore Water and Waste Management (LWWM) Project”, which has the following two components.

- i. Construction of Surface Water Treatment Plant (SWTP) at BRBD Canal, Near Ravi Syphon, Lahore and
- ii. Laying of Sewerage System from Larech Colony to Gulshan-e-Ravi, Lahore (Through Trenchless Technology).”

A joint venture of four engineering firms, EBA, ECSP, EEC and TEAM has been awarded the consultancy services for acting as ‘Owner’s Engineer and Project Management Consultant (PMC)’ for the second project, which covers Central Zones of Lahore, the most thickly populated and congested with respect to others zones. This project is aimed to provide the engineering consultancy services by a JV so that construction of trunk sewers through micro-tunneling/trenchless technology, for collecting the sewage discharge from several sewage lift stations, which are presently pumping the sewage effluent into the existing natural and man-made storm water drains. The proposed trunk sewers will convey the sewage effluent to the proposed disposal station at the Bund Road near the existing Gulshan-e-Ravi Sewage Disposal Station.

The economic evaluation is conducted so as to establish the economic justification for the improvement-construction of proposed trunk sewerage. The economic analysis has been undertaken with a view of compute economic indicators required for decision making with regard to project implementation. Such economic analysis consists of:

- Estimation of project economic benefits and costs
- Derivation of economic indicators and

2. BROAD CRITERIA

In the economic-Economic analysis, following broad parameters have been adopted:

- Project Cost (Total) Rs. 49,270 million
- Project Cost (Economic Purposes) Rs. 43,357 million
(Cost towards consulting services, etc. excluded)
- Economic Factor 0.88
- O&M Cost Including Power Bills (First Year) 0.42 % of Project Cost
(Determined on the basis of proposed Lift Stations & project Cost)
- Economic Life 25 Years
- Salvage Value 50 %
- Discount Rate 10%

3. PROJECT COSTS

The project cost estimates have been determined on the basis of preliminary designs and work out as Rs. 49,270 million, the summary of cost is included in Table - 1.1

Table - 1.1 Summary of Project Cost

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT
LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE
ABSTRACT OF COST

Exchange Rate 3-May-24

USD 278.25

EURO 300.56

Sr. No.	DESCRIPTION	AMOUNT (PKR)
A	Sewer Line from Karachi Phatak to Qartaba Chowk	8,918,781,371
B	Sewer Line from Gurumangat to Qartaba Chowk	7,069,359,000
C	Combined Reach (Toheed Park To Gulshan - E - Ravi)	8,277,031,033
D	Sheet Piling & Bracing System	1,530,502,352
E	Connections with Existing Disposal Stations to be eliminated	2,520,066,565
F	Construction of Disposal Station of 688 Cusecs Capacity at Gulshan-e-Ravi (Ultimate Capacity = 848 Cusecs)	8,503,792,970
G	Rehabilitation of Outfall Channel	395,218,880
H	Rider Sewers & Bricklining of Cantt. & Gulberg Drains	861,406,423
I	Inlet Screens at all Stations with Odor Control System, Drop Manholes	1,369,346,955
J	Site Facilities	292,950,000
K	Miscellaneous	
L	450 kW Solar System	80,000,000
	Sub-Total (A+B+C+D+E+F+G+H+I+J+K+L) (Rs.)	39,818,455,548
	TOTAL	39,818,455,548
	PRA @ 5% (Counterpart Funding)	1,990,922,777
	GRAND TOTAL	41,809,378,325
	Duties & Taxes on Import Items	1,860,620,000
i)	Cost for Owner's Engineer Consultancy Services	1,274,380,658
ii)	ESMP including Livelihood Restoration Plan (LRP) & Resettlement plan, Health & Safety Cost	191,007,157
iii)	Provision for Electrical connecting charges at D/station/Grid Station	1,515,745,400
iv)	PMU-L-WASA Establishment	296,050,000
v)	Shifting of Utility services @ 1 % of Project Cost	398,184,555
vi)	Provision for Road Restoration / Remodeling & Bridge Construction @ 1 %	398,184,555

vii)	Contingencies (excluding land acquisition and price escalation)	242,673,577
ix)	Provision for Media Campaign @ 0.05 %	
x)	Provision of Horticulture @ 0.05 %	
xi)	Price Adjustment/ Variation & Escalation @ 2 % (As per FIDIC Guidelines)	796,369,111
xii)	01 Years O&M	487,809,279
	Sub-total For Provisional Sums from item (i to xii)	7,461,024,293
Total Amount (Rs.).		49,270,402,618
Total Amount in US\$ (In Million) @ 1US\$ = Rs 278.25 (SBP Exchange Rate May 03, 2024)		177,072,426

For the purposes of economic analysis, the base cost of Rs. 49,270 million is determined as above and has been adopted.

Financial costs conversion to into economic costs, a factor of 0.88 has been adopted. This factor ensures that elements of indirect or hidden part of all externalities, if present in the project cost are removed.

4. PROJECT BENEFITS

Benefits attributed to the implementation of the project can be classified as 'tangible' or direct and 'intangible' or indirect benefits. Tangible benefits are those which can be quantified and to which a monetary value can be assigned whereas intangible benefits though socially desirable cannot be quantified in monetary terms and are usually described qualitatively.

4.1 Tangible Benefits

The benefits out of provision of sanitation-sewerage possibly determined in physical terms. These have been explained in later section

4.2 In-tangible Benefits

In-tangible benefits comprise of a number of economic and social advantages which though important, cannot be quantified in monetary terms. These include:

- Health and control of environmental pollution.
- Safety of public and private property.
- Safety of roads infrastructure.
- Convenience of workers call towards punctuality in reaching offices-work places and return to homes.

- Prevention of communicable and water borne diseases thereby resulting in healthy people who deliver outputs more effectively as such adding to economic development.
- Others.

In the economic analysis, only tangible benefits have been considered.

4.3 Salvage Values

The following salvage values (at the end of the economic life of the proposed sewerage) have been allowed at the end of the economic life of each of the component of the project, Table – 1.3

Table 1.3, Salvage Value at the end of Service Life

No.	Project Component	Residual Value (%)
1.	Earthwork	0%
2.	Pipe and Shafts	50%
3.	Structures	50%

5 RECURRENT COSTS

WASA is already providing operation and maintenance services in the service area which will be utilized for the operation and maintenance purposes. However, for the purposes of economic evaluation, the Operating and Maintenance (O&M) costs are determined on the basis of share of various components of proposed lift stations and is included in Table – 1.4.

Table – 1.4, Determination of O&M Cost of the Project

Sr. No.	Description	O&M Cost Rs. In millions	Share of Components based on Existing O&M
1	Electricity	223.562	99.59%
2	Establishment Charges	0.600	0.267%
3	Repair, etc.	0.300	0.133%
	Total:	224.463	100%
	O&M, % age of Cost	0.42 %	

It is noted from the above table that O&M cost works out as 0.92% of the total project cost.

6 TARIFF STUDY

6.1 Existing Tariff

The structure of present tariff for sewer tariff is as under:

- Domestic
- Commercial

The present structure of Tariff Water and Sewerage) is included in Table – 1.5.

Table -1.5, Existing Tariff by WASA

Domestic

DOMESTIC TARIFF RATES	
Area of Plot	W&S Rate For 30-Days (PKR)
Up to 3 Marla	427
3.1 to 5	610
5.1 to 7	1,830
7.1 to 10	2,440
10.1 to 15	3,050
15.1 to 20	4,880
20.1 to 40	6,100
40.1 & Above	7,320

Commercial

COMMERCIAL TARIFF RATES	
Area of Plot	W&S Rate For 30-Days (PKR)
Commercial-A1	1,220
Commercial-A	3,975
Commercial-B	6,521
Commercial-C	9,071
Commercial-D	12,755
Commercial-E	16,443
Commercial-F	23,812
Commercial-G	31,183
Commercial-H	38,556

6.2 Future Sewerage Tariff Level

Proposed tariff has been assessed on the basis of the following steps:

- Assessing Economic sustainability
- Assessing beneficiaries willingness to pay

a. Anticipated Sustainable Tariff-Domestic

The sustainable tariff on the basis of monthly income of the beneficiaries and affordability level is worked out and included in Table - 1.6.

Table – 1.6, Sustainable Sewerage Tariff

DOMESTIC TARIFF RATES		
Area of Plot	W&S Rate For 30-Days (PKR)	W&S Rate For 30-Days (PKR) in 2050
Up to 3 Marla	427	2,604
3.1 to 5	610	3,720
5.1 to 7	1,830	11,160
7.1 to 10	2,440	14,880
10.1 to 15	3,050	18,600
15.1 to 20	4,880	29,760
20.1 to 40	6,100	37,200
40.1 & Above	7,320	44,640

b. Sustainable Tariff-Non Domestic

In the absence of WASA's record about growth pattern of commercial connections, it is assumed that the growth trends will follow the same trend to that of the domestic connections. The industrial connections are to be discouraged; industries should make their own arrangements for waste water disposal.

c. Connections Fee

When viewed in the context of cost of material, labour, etc. the installation / connection fee, it is proposed to increase by 7.5 % after every year.

d. Varying Tariff

With the increase in cost of facilities and raise in socioeconomic level, the tariff should vary over the period; an annual increase of 7.5% is proposed and taken into consideration in this analysis.

7. Tangible Benefits

7.1 Savings due to Closure of Pumping Stations

The project covers a total length of 20 km and 32 pumping stations becomes redundant and nonfunctional.

a) Savings in Energy Bill

The project covers a total length of 20 km and 32 pumping stations becomes nonfunctional. Projected savings in annual electricity cost is given in Table - 1.7.

Table – 1.7, Electricity Cost Saving

Year	Electricity Cost Saving (Rs in Millions)
2025	300.000
2030	430.668
2035	618.309
2040	887.663
2045	1,274.355
2050	1,829.501

b) Savings in Labour Cost

The project covers a total length of 20 km and 32 pumping stations becomes nonfunctional thus projected estimated saving in labour cost is as under Table - 1.8.

Table – 1.8, Labour Cost Saving

Year	Labour Cost Saving (Rs in Millions)
2025	211.600
2030	303.779
2035	436.114
2040	626.098
2045	898.845
2050	1,290.408

c) Savings in Maintenance Cost

The project covers a total length of 20 km and 32 pumping stations becomes nonfunctional thus projected estimated savings in maintenance cost of 32 pumping stations is as under Table - 1.9.

Table – 1.9, Maintenance Cost Saving

Year	Maintenance Cost Saving (Rs in Millions)
2025	28.590
2030	41.045
2035	58.926
2040	84.596
2045	121.449
2050	174.356

d) Savings in Generator Cost

Each pumping station is equipped with standby generator facility. After completion of this project these 32 standby generator will become nonfunctional. Projected savings in annual savings in annual energy bills is given in Table – 1.10.

Table – 1.10, Generator Cost Saving

Year	Generator Cost Saving (Rs in Millions)
2025	160.162
2030	229.933
2035	330.098
2040	473.899
2045	680.344
2050	976.722

e) Savings in Machinery Cost

Heavy machinery like Suckers etc is being used by wasa which will become nonfunctional. Projected savings in annual machinery cost is given in Table – 1.11.

Table – 1.11, Machinery Cost Saving

Year	Machinery Cost Saving (Rs in Millions)
2025	96.251
2030	138.180
2035	198.375
2040	284.793
2045	408.858
2050	586.968

7.2 Savings in Desilting Expenditure

WASA, Lahore incurs a heavy expenditure on desilting of drains annually. Projected savings in annual desilting expenditure is given in Table - 1.12.

Table – 1.12, Desilting Cost Saving

Year	Desilting Cost Saving (Rs in Millions)
2025	224.000
2030	321.580
2035	461.671
2040	662.788
2045	951.518
2050	1,366.028

7.3 Savings in Annual Repairs of Road Infrastructure

According to TEPA, heavy rains and storm in monsoon season badly affects road infrastructure of Lahore. In 2019, TEPA spend Rs. 155 Million on account of repairs of roads. Projected savings in annual repairs in road infrastructure is given in Table - 1.13.

Table – 1.13, Cost Saving in Annual Repairs of Road Infrastructure

Year	Cost Saving in Annual Repairs of Road Infrastructure (Rs in Millions)
2025	207.475
2030	297.857
2035	427.612
2040	613.893
2045	881.322
2050	1,265.253

7.4 Savings in Health Expenditure

According to a study conducted by ADB, Pakistan spends 3.94% of the GDP on health due to poor sanitation conditions. Considering the same amount as assessed by World Bank, in Pakistan, per capita Sanitation related expenditure on Health is Rs. 18,290/-, which is computed as in Table - 1.14.

Table – 1.14, Savings in Health Expenditure

Pakistan GDP 2021 (USD Million)	346,340.000
Cost of Poor Sanitation % of GDP	3.94%
Cost of Poor Sanitation (USD Million)	13,645.796
Exchange Rate 1 USD to PKR (as on 1st November 2022)	278.502
Cost of Poor Sanitation (PKR Million)	3,8800,381.478
Population as per 2017 census (Million)	207.774
Per Capita Cost of Poor Sanitation (PKR)	18,290.939

7.4 Savings in Repair & Maintenance of Sewers and Drains

The project covers a total length of 20 km and 32 pumping stations becomes nonfunctional. Projected savings in annual repairs & maintenance of sewers & drains is given in Table – 1.15.

Table – 1.15, Savings in Repair & Maintenance of Sewers and Drains

Year	Savings in Repair & Maintenance of Sewers (Rs. in Millions)	Savings in Repair & Maintenance of Drains (Rs. in Millions)
2025	1.650	4.500
2030	2.368	6.460
2035	3.400	9.274
2040	4.882	13.314
2045	7.008	19.115
2050	10.062	27.442

Based on these assumptions annual stream of revenue and benefits generated from the project is listed in Table - 1.16

1.16 Annual Stream of Project Benefits

ANNUAL STREAM OF BENEFITS																			Million Rs.	

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8. Economic Analysis

The Projected stream of economic benefits over the economic life of the project has been compared to the estimated stream of economic costs by bringing the two streams to a uniform basis through the process of discounting. To judge the economic viability of the project, the following economic indicators have been computed:

- Net Present Value (NPV).
- Benefit Cost Ratios (B/C Ratio).
- Economic Internal Rate of Return (EIRR)

Results of economic analysis for each component is summarized in Table – 1.17

Table – 1.17 Analysis Results

	Economics Parameters	10% Discount Rate
1	Present value of the benefits- (In Rs. million)	70,294.333
2	Present value of the costs- (In Rs. million)	41,287.943
3	B/C Ratio	1.5478
4	EIRR(%)	15%

9. Sensitivity Analysis

Sensitivity analysis is particularly concerned with factors and combinations of factors that may lead to unfavorable consequences. These factors normally identified in the project (logical) framework as „project risks” or “project assumptions”. Sensitivity analysis tries to estimate the effect of achieving project objectives if certain assumptions do not, or only partly, occur.

9.1 The Purpose of Sensitivity Analysis

Sensitivity analysis is a technique for investigating the impact of changes in project variables on the base-case (most probable outcome scenario). Typically, only adverse changes are considered in sensitivity analysis. The purpose of sensitivity analysis is:

1. to help identify the key variables which influence the project cost and benefit streams
2. to investigate the consequences of likely adverse changes in these key variables
3. to assess whether project decisions are likely to be affected by such changes
4. to identify actions that could mitigate possible adverse effects on the project.

In order to examine the effect of increase or decrease in project cost or economic benefits of project, three alternative economic analyses have been undertaken on following basis;

1. Project benefits estimated for the 'base case' turn out to be 10 percent lower than originally estimated with no change in project costs.
2. Project economic costs over-run by 10 percent than those estimated for the 'base case' with no change in project costs.
3. Project benefits turn out to be lower along with cost over-run, each by 10 percent.

The result of the sensitivity analysis under the above-mentioned assumptions are summarized in the Table 1.18 below.

Table – 1.18 Sensitivity Analysis Results

Assumptions	EIRR (Present)
Base case	15%
Sensitivity Tests	
i. Benefits - 10%	13.69%
ii. Costs +10%	13.81%
iii. Benefit - 10% & Costs +10%	12.56%

The results of the above mentioned sensitivity analysis clearly depict that in all cases the project is not highly sensitive to any of the above mentioned assumptions and EIRR of the project is well within the reasonable range.

Annexure 1

CALCULATION OF ECONOMIC INTERNAL RATE OF RETURN (EIRR)										
(In Millions)										
YEAR S	n	Year of Construction	Year after Construction n	PROJECT ECONOMIC COSTS	PROJECT ECONOMIC BENEFITS	NET BENEFITS	Project Sensitivity Test			
							Base Case (A)	Benefits Down 10% (B)	Cost Up 10% (C)	B & C Simultaneously
2024		1		21,678.977		-21,678.977	-21,678.977	-21,678.977	-23,846.875	-23,846.875
2025	1	2		21,678.977	0.000	-21,678.977	-21,678.977	-21,678.977	-23,846.875	-23,846.875
2026	2		1	197.527	5,521.305	5,323.778	5,323.778	4,771.647	5,304.025	4,751.894
2027	3		2	201.989	5,783.078	5,581.089	5,581.089	5,002.781	5,560.890	4,982.583
2028	4		3	244.050	6,020.835	5,776.785	5,776.785	5,174.701	5,752.380	5,150.296
2029	5		4	294.899	6,273.105	5,978.206	5,978.206	5,350.895	5,948.716	5,321.406
2030	6		5	300.401	6,539.501	6,239.100	6,239.100	5,585.150	6,209.060	5,555.110
2031	7		6	323.788	6,821.019	6,497.231	6,497.231	5,815.129	6,464.852	5,782.750
2032	8		7	348.999	7,118.731	6,769.733	6,769.733	6,057.859	6,734.833	6,022.960
2033	9		8	376.176	7,433.798	7,057.622	7,057.622	6,314.243	7,020.005	6,276.625
2034	10		9	401.632	7,767.473	7,365.781	7,365.781	6,589.034	7,325.612	6,548.865
2035	11		10	437.889	8,121.110	7,683.221	7,683.221	6,871.110	7,639.432	6,827.321
2036	12		11	463.194	8,496.174	8,032.979	8,032.979	7,183.362	7,986.660	7,137.043
2037	13		12	489.963	8,894.249	8,404.285	8,404.285	7,514.860	8,355.289	7,465.864
2038	14		13	518.281	9,317.048	8,798.767	8,798.767	7,867.062	8,746.939	7,815.234
2039	15		14	548.237	9,766.428	9,218.191	9,218.191	8,241.548	9,163.367	8,186.724
2040	16		15	574.428	10,244.395	9,669.967	9,669.967	8,645.527	9,612.524	8,588.085
2041	17		16	636.519	10,753.124	10,116.606	10,116.606	9,041.293	10,052.954	8,977.641
2042	18		17	705.329	11,294.971	10,589.642	10,589.642	9,460.145	10,519.109	9,389.612
2043	19		18	781.586	11,872.486	11,090.899	11,090.899	9,903.651	11,012.741	9,825.432
2044	20		19	866.097	12,488.435	11,622.338	11,622.338	10,373.434	11,535.728	10,286.885
2045	21		20	1,042.875	13,145.816	12,102.941	12,102.941	10,788.360	11,998.654	10,684.072
2046	22		21	1,134.675	13,847.880	12,713.205	12,713.205	11,328.417	12,599.737	11,214.949
2047	23		22	1,234.561	14,598.150	13,363.589	13,363.589	11,903.774	13,240.133	11,780.318
2048	24		23	1,343.246	15,400.452	14,057.206	14,057.206	12,517.161	13,922.881	12,382.836
2049	25		24	1,461.505	16,258.935	14,797.430	14,797.430	13,171.537	14,651.280	13,025.386
2050	26		25	1,566.832	17,178.102	15,611.270	15,611.270	13,893.460	15,454.587	13,736.777
2051	27		26	1,704.785	18,156.488	16,451.703	16,451.703	14,636.054	16,281.224	14,465.575
2052	28		27	1,854.892	19,205.412	17,350.521	17,350.521	15,429.979	17,165.031	15,244.430
2053	29		28	2,018.221	20,330.625	18,312.405	18,312.405	16,279.342	18,110.583	16,077.520
Total				65,430.589	308,649.124	243,218.535	243,218.535	212,353.623	236,675.476	205,810.564
Discount Rates				Present Worth		NPV (Base Case)	NPV (Base Case)	NPV (Benefits Down 10%)	NPV (Cost Up 10%)	NPV (Both)
		5%		48,507.271	134,375.402	79,469.301	79,469.301	66,671.644	74,618.574	61,820.917
		10%		41,287.943	70,294.383	22,616.041	22,616.041	16,225.643	18,487.247	12,096.849
		15%		37,181.054	42,754.656	-3.092	-3.092	-3,720.888	-3,721.198	-7,438.994
		20%		34,293.783	29,069.778	-10,068.968	-10,068.968	-12,491.450	-13,498.347	-15,920.828
		25%		32,002.438	21,368.172	-14,907.900	-14,907.900	-16,617.354	-18,108.144	-19,817.598
		30%		30,067.047	16,579.555	-17,313.543	-17,313.543	-18,588.894	-20,320.248	-21,595.598
Internal Rate of Return						15.00%	15.00%	13.6%	13.81%	12.56%
B/C Ratio @										
		5%				2.6383	2.6383	2.3745	2.5383	2.2745
		10%				1.5478	1.5478	1.3930	1.4478	1.2930
		15%				0.9999	0.9999	0.8999	0.8999	0.7999
		20%				0.7064	0.7064	0.6358	0.6064	0.5358
		25%				0.5342	0.5342	0.4807	0.4342	0.3807
		30%				0.4242	0.4242	0.3818	0.3242	0.2818

12. IMPLEMENTATION SCHEDULE

12.1 IMPLEMENTATION SCHEDULE/GANTT CHART:

Project of Laying of Sewerage System from LARECH Colony to Gulshan-e-Ravi through Trenchless Technology already stands approved with Gestation period of 30 Months. Similarly, instant 2nd revision of same project is being proposed with implementation duration of 30 months on EPC mode. Project Management Consultancy services has already been hired for project with total duration of 36 Months (6 Months Design Phase+ 30 Months Execution Phase)

12.2 RESULT BASED MONITORING (RBM) INDICATORS:

RBM indicators matrix

RESULT BASED MONITORING INDICATORS

Input	Output	Outcome		Targeted Impact
		Base line indicator	Targets after Project Completion	
Rs. 49,270.40/- Millions	i) Execution of 32.369 km Sewer line through Trenchless technology ii) Construction of new 848 Cfs Disposal Station. at Gulshan-e-Ravi.	Monitoring and Evaluation of execution of work by: i) Project Management Unit ii) AIIB Appraisal team iii) Project Management Consultant iv) Three Tier GRM establishment	Successful Implementation of LARECHS TBM Project	i) Redressal of sewerage related issues in Central Zone ii) Restoration of natural ecology and recreational activities along cantonment drain of Lahore iii) Mitigation of storm runoff in chronic areas of central zone

12.3 IMPLEMENTATION PLAN:

Implementation Plan for Package 1 and Package of Instant project are detailed below

ID		Task Mod.	Task Name	Duration	Start	Finish	Predecessors	1st Half Qtr 1	Qtr 2	2nd Half Qtr 3	Qtr 4	1st Half Qtr 1	Qtr 2	2nd Half Qtr 3	Qtr 4	1st Half Qtr 1	Qtr 2	2nd Half Qtr 3	Qtr 4
1			Sewerage System from Larech Colony to Gulshan-e-Ravi, Lahore (Through Trench-Less Technology)	652 days	Wed 28/01/26	Sun 01/10/28													
2			Issuance of Notification of Commencement	7 days	Wed 28/01/26	Sat 07/02/26													
3			Detailed Field Surveys including Geotechnical & Topographical Surveys	40 days	Sat 07/02/26	Wed 08/04/26	2												
4			Detailed Design Report Preparation & Submission	60 days	Wed 08/04/26	Tue 07/07/26	3												
5			Site Mobilization	80 days	Sat 07/02/26	Sun 07/06/26	2												
6			Procurement of Equipment	140 days	Tue 07/07/26	Tue 02/02/27													
7			Microtunneling Machines	140 days	Tue 07/07/26	Tue 02/02/27	4												
8			Inlet Screens	100 days	Thu 06/08/26	Sun 03/01/27	4FS+20 days												
9			Sheet Piles	80 days	Fri 21/08/26	Sat 19/12/26	4FS+30 days												
10			Equipment for Drop Manholes	80 days	Thu 06/08/26	Fri 04/12/26	4FS+20 days												
11			Instrumentation	80 days	Fri 21/08/26	Sat 19/12/26	4FS+30 days												
12			SCADA & Accessories	100 days	Fri 21/08/26	Mon 18/01/27	4FS+30 days												
13			Site works	580 days	Mon 09/03/26	Wed 26/07/28													
14			High Strength RCC Pipe Manufacturing	300 days	Mon 09/03/26	Wed 02/06/27	5SS+20 days												
15			Sheet Piles Fixing & Dismantling in Launching & Receiving Pits	350 days	Sat 19/12/26	Sat 27/05/28	9												
16			Excavation Works For Pits	300 days	Sun 03/01/27	Tue 28/03/28	15SS+10 days												
17			RCC Works in Pits	300 days	Mon 18/01/27	Wed 12/04/28	16SS+10 days												
18			Civil Works For Drop Manholes	150 days	Sun 14/11/27	Mon 26/06/28	17SS+200 days												
19			Pipe Jacking For 1500 mm Lines	300 days	Tue 02/02/27	Thu 27/04/28	17SS+4.67 days,7,												
20			Pipe Jacking For 2400 mm Lines	300 days	Tue 02/02/27	Thu 27/04/28	17SS+10 days,7												
21			Pipe Jacking For 3500 mm Lines	250 days	Fri 19/03/27	Tue 28/03/28	17SS+40 days												
22			Backfilling of Pits	300 days	Thu 04/03/27	Sat 27/05/28	17SS+30 days												
23			Installation of Electrical Equipment, Instrumentation, Cabling	20 days	Mon 26/06/28	Wed 26/07/28	18,19,20,21												
24			Testing & Commissioning of Complete System	45 days	Wed 26/07/28	Sun 01/10/28	23												

Project: Implementation Schedu
Date: Wed 30/10/24

- Task

Split

Milestone

Summary
-
- Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

-
- Manual Task

Duration-only

Manual Summary Rollup

Manual Summary
-
- Start-only

Finish-only

External Tasks

External Milestone

-
- Deadline

Progress

Manual Progress
-

ID		Task Mod.	Task Name	Duration	Start	Finish	Predecessors	01 January21/12 21 May10/05 11 October27/09 01 March14/02 21 July04/07 11 December21/11 01 May30/01 01 May09/04 21 September18/06 21 September27/08															
1			Gulshan-E-Ravi Disposal Station (Package 2)	607 days	Wed 28/01/26	Sun 26/09/27																	
2			Issuance of Notification of Commencement	7 days	Wed 28/01/26	Tue 03/02/26																	
3			Detailed Field Surveys including Geotechnical & Topographical Surveys	30 days	Wed 04/02/26	Thu 05/03/26	2																
4			Detailed Design Report Preparation & Submission	45 days	Fri 06/03/26	Sun 19/04/26	3																
5			Site Mobilization	15 days	Wed 04/02/26	Wed 18/02/26	2																
6			Procurement of Equipment	240 days	Mon 20/04/26	Tue 15/12/26																	
7			Disposal Pumps	240 days	Mon 20/04/26	Tue 15/12/26	4																
8			Inlet Screens	165 days	Mon 20/04/26	Thu 01/10/26	4																
9			Sheet Piles	150 days	Mon 20/04/26	Wed 16/09/26	4																
10			Instrumentation	135 days	Mon 20/04/26	Tue 01/09/26	4																
11			Electrical Panels	210 days	Mon 20/04/26	Sun 15/11/26	4																
12			SCADA & Accessories	200 days	Mon 20/04/26	Thu 05/11/26	4																
13			Transformers & Gensets	180 days	Mon 20/04/26	Fri 16/10/26	4																
14			Site works	555 days	Thu 19/02/26	Fri 27/08/27																	
15			Civil Works For Disposal Station	300 days	Thu 19/02/26	Tue 15/12/26	5																
16			Rehabilitation of Outfall Channel	225 days	Sat 30/05/26	Sat 09/01/27	15SS+100 days																
17			Installation of Mechanical Equipment in Disposal Station	150 days	Wed 16/12/26	Fri 14/05/27	15,7,8																
18			Installation of Electrical Equipment, Instrumentation, Cabling	120 days	Sun 14/02/27	Sun 13/06/27	11,12,13,15FS+60 days,10																
19			PLC & SCADA Programming	75 days	Mon 14/06/27	Fri 27/08/27	18																
20			Testing & Commissioning of Complete System	30 days	Sat 28/08/27	Sun 26/09/27	19																

Project: Implementation Schedule
Date: Wed 30/10/24

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

Page 1

12.4 M&E PLAN:

Monitoring & Evaluation Plan

MONITORING & EVALUATION PLAN

Monitoring is the routine to collect data and information on activities and operations (e.g. from customers, markets etc.). Evaluation is then used to analyze the compiled data and to compare the actual results of a business or project against the set strategic plans. Monitoring and Evaluation (M&E) helps government/monitoring entity to track how effective and efficient resources are used and whether one is on track to meet its set goals. This tool provides information on how a effective sanitation system can implement and improve its monitoring and evaluation scheme. Hands-on insights are provided through an overview of M&E activities of any project.

Evaluation is used to compare actual project results against strategic plans. Unlike monitoring, evaluation is not done on a routine basis but done at specific times (i.e. monthly, bi-yearly, yearly etc.). Evaluation refers to a range of activities that are undertaken to determine if the project activities had the desired impact. There are several types of evaluations that may be appropriate for social businesses: formative evaluations, outcome evaluations, impact.

Monitoring and evaluation (M&E) are important for projects relating to sewerage system in several ways:

- An M&E system helps the monitoring entity to understand the public needs, to inform the management and adapt plan/strategies accordingly where needed.
- A well-established M&E system can also help the monitoring entity to determine whether he/she is accomplishing the desired social/health impact. This includes monitoring effective regarding effective collection and disposal of sewage waste in case of large scale sewerage projects.

WASA is planning to establish dedicated Project Management Unit for instant project to monitor and evaluate the project during design and execution phase of project, similarly dedicated Project Management Consultant (PMC) for project is also hired by WASA for review/update refinement of design prepared earlier by NESPAK, besides design phase, consultant will also carryout Resident Supervision and Project Management support. In order to facilitate monitoring and evaluation process, entire Project is subdivided into two packages:

Sr. No	Description of Contracts
1	Laying of Sewer Line from Karachi Phatak and Gurumangat Gulberg to Gulshan-e-Ravi (through trenchless technology) and connections with existing disposal stations to be eliminated
2	Construction of Ultimate Disposal Station of 848 cfs Capacity at Gulshan-e-Ravi and rehabilitation of outfall channel including Allied Works.

During Project execution phase, consultant will also monitor the performance of contractors for smooth functioning of scheme. This will also include chalking out Monitoring and Evaluation plan for evaluating the performance/progress of contractor on fortnightly basis during course of work as well as formulation of progress reports for furnishing to PMU/WASA for keeping eye on project milestones. Similarly, Financing Agency (AIIB) Monitoring Team will also monitor progress of project during course of execution.

A three Tier Grievance Redressal Mechanism will also be established to redress the complaints of applicants, if lodged by applicants during course of work

Operational Monitoring

To better facilitate the management, operation and control of the new facilities i.e. disposal pumping station at Gulshan-e-Ravi and trunk sewerage lines, to be constructed under this project, a state-of-the-art computerized control and monitoring system termed as 'Supervisory Control and Data Acquisition' (SCADA) system shall be designed, supplied, installed and commissioned by the Contractor as per the design described hereunder.

The **key objectives** of SCADA system shall be as follows:

- i. Centralized and live monitoring of the new facilities i.e., disposal station at Gulshan-e- Ravi and trunk sewer lines
- ii. Streamlining the pump operation and control at disposal pumping station at Gulshan-e-Ravi
- iii. Monitoring of generators at disposal pumping station at Gulshan-e-Ravi
- iv. Measurement of level, flow and presence of sewage gasses at selected locations in the trunk sewer lines
- v. Availability of record/data log and reports from the SCADA system

SCOPE OF WORK

- a) Product Life for the SCADA System technology is envisaged to be more than 20 years and reliability of the products shall be of highest level having international references/experience of past 5 years at least.
- b) SCADA system shall consist of various hardware and software components. SCADA Software shall include functions of Human Machine Interfaces (Operator PCs or Servers with graphics/mimics), Logging, Alarm Displays, Historian, Daily/ Monthly Reporting and video streaming/recording, and whereas SCADA hardware shall mainly consist of RTUS/PLCs/Radio Modems, Flow/Level Instrumentation, gas detection sensors, sewage video cameras, cables, panels, uninterruptable power supply (UPS) having backup of 6 Hours at various locations. A conceptual block diagram is shown below **Figure 4**
- c) SCADA system shall carry out complete monitoring and control of all the pumps to be installed at the new disposal station.
- d) SCADA system for the new Disposal Pumping Station Gulshan Ravi shall also monitor conditions of the trunk sewerage lines to be furnished with local RTUs and necessary instruments/sensors i.e. Level/Flow, Gas detection and video, etc.
- e) Status and control of the associated Generators/ATS shall also be provided at the SCADA which would include minimal functions i.e. Run Status, Start/Stop, major parameters of power, etc. of the generators.
- f) SCADA Centre shall also be developed and installed at WASA Head Office which shall be linked to the SCADA Centre at Disposal Station at Gulshan Ravi through a Fibre Optic leased line/Broadband Internet connection.
- g) The scope also includes training of WASA staff for the system operation & maintenance, necessary tools, test and spares for the system operation and maintenance. Contractor shall arrange separate spares for the commissioning purposes

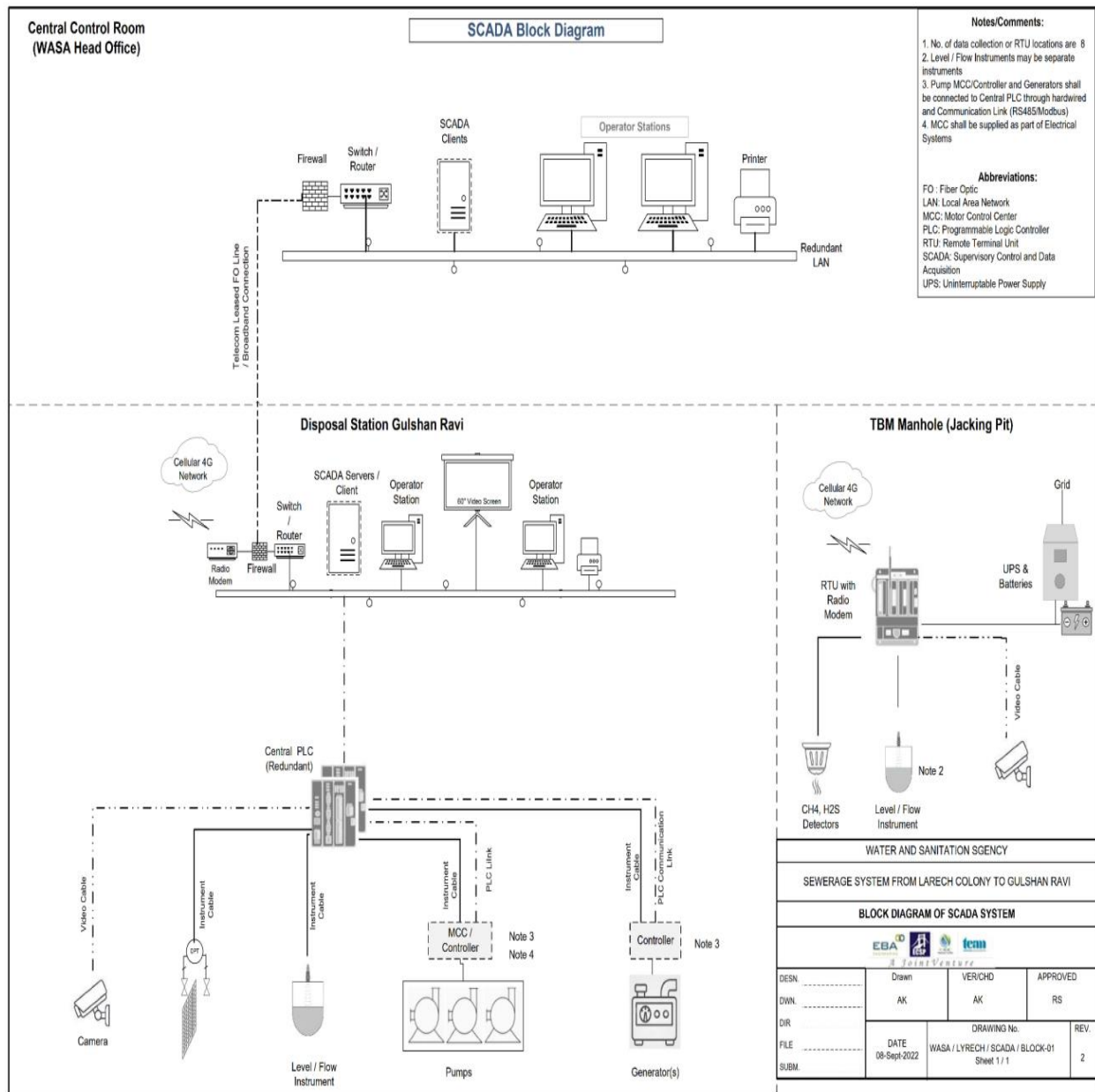
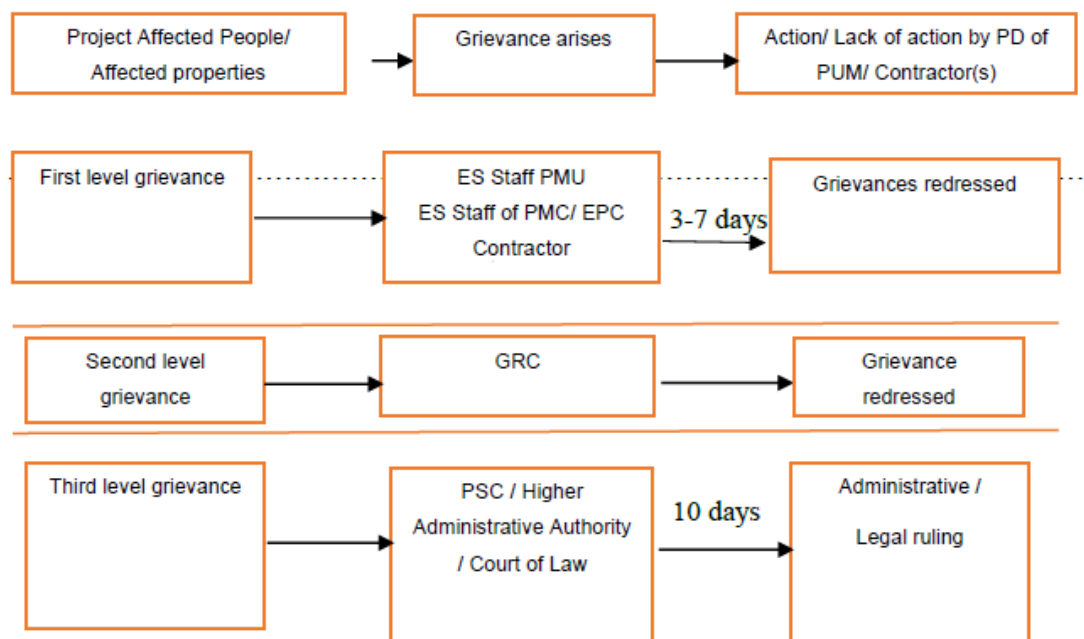


Figure 4: Block Diagram of SCADA System

THREE TIER GRIEVANCE REDRESSAL MECHANISM

In order to ensure effective Monitoring and implementation of instant Scheme followed by disbursement of Cost to Affected people concerning to Land and Non Land Assets, it has been proposed to hire services of third party monitoring consultant, who will be responsible for effective disbursement of compensation cost to affected people and subsequent training of affected person for their skill development for starting any other suitable business for earning as per interest and need. An independent Project Management Unit for mega project of LARECHS Trunk sewer has been established for which Remuneration structure and JDs have already been approved by PDWP in its 18th Meeting held on 27.09.2022.

PMU will work in close liaison with Third Party Consultant regarding various matters relating to Disbursement of Costs in form of LRP to affected people of LARECHS Trunk Sewer Project. Further in addition to above, Continuous Monitoring will also be done throughout Project Duration by PMU staff and Bank Environmental and Social Safeguard Specialists. Two Tier GRM have been established to redress any concerns of the project. Third Tier of GRM is in the formulation stage. A Project specific three-tier GRM designed to provide a time-bound, early, transparent and fair resolution for AP (Affected People) and stakeholder grievances. The Manager (Environmental and Social) of PMU will undertake public awareness campaigns on the GRM with the assistance of Environmental and Social Specialist (ESS) of PMC and EPC/DBO Contractor. All complaints received verbally or in writing will be properly documented and recorded in the Complaint Management Register(s).



THREE TIER GRM SCHEMATIC DIAGRAM

12.5 RISK MITIGATION PLAN:

Risk Mitigation Matrix

RISK MITIGATION PLAN

LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN E RAVI						
Sr. No	Risks	Causes/Factors	Effects	Risk Mitigation Plan	Institution/s Responsible to take corrective measures to mitigate Risk	Nature of Risk
1	Financial Risks					
	Lack of funds for Project Execution	No funds from the AIIB	Time delays, project cost will increase	Approach AIIB (or provincial govt may seek funding from Federal Govt)	Project Management unit (PMU)/WASA Lahore	Low
	Insufficient funding for O&M	Provincial government does not have funds (or the relevant authority / community does not have sufficient funds)	Disposal Station shut down which may cause accumulation of sewerage in the Trunk Sewers	The operation mechanism would be formulated to reduce OPEX cost through enhanced efficiency and maintenance of pumps.	PMU/ WASA Lahore	Medium
	Tariff revision to finance the Trunk Sewer O&M expenses	Increase in tariff will trigger political risk.	May trigger protest from the general public	Tariff may be increased in phased manner rather than rapid/sharp tariff increase by perusing government entities.	WASA, Lahore /PMU	High
	HR Risks					

2	Lack of Technical Personals	No previous expertise of carrying out Trenchless technology available with WASA, Lahore	Ineffective and inefficient management resulting in Cost overruns and delays	Dedicated PMU staff comprising of seasoned officers will be hired to monitor & evaluate project	PMU/ WASA Lahore	Low
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LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN E RAVI						
Sr. No	Risks	Causes/Factors	Effects	Risk Mitigation Plan	Institution/s Responsible to take corrective measures to mitigate Risk	Nature of Risk
3	Technology Risks					
	Suitable selection of Technology and TBM Machine	Fault in design or poor assessment of geological stratigraphy	Expensive technology with difficult operations and imported spare parts will be used which can result in financial and operational implications/difficulties	Detailed geotechnical investigation, and comprehensive formulation of design alongwith Project management Consultant before detailed design by EPC Contractor	Project Management Consultant (PMC), PMU, EPC Contractor	High
4	Continuous supply of Energy	Non-availability of continuous energy supply will impact the existing energy deficit	Hindrance will result in project delays and cost overruns	Ensure continuous supply of energy through exploring alternate energy sources such dedicated Grid station and Genset etc.	PMU, PMC, EPC Contractor	High
5	Legal/Regulatory Risks					
	Legal, Policy and Regulatory framework	Obtaining NOCs from relevant government departments/authorities	Legal issues may halt project at any stage of the project life cycle	NOCs to be obtained from departments such as LDA, Railways etc., and/or any other relevant department at the planning stage of the project.	PMU	Medium

LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN E RAVI						
Sr. No	Risks	Causes/Factors	Effects	Risk Mitigation Plan	Institution/s Responsible to take corrective measures to mitigate Risk	Nature of Risk
6	Political Risks	Revision of tariff will be a challenge keeping in view public sentiment and political interests	Project delay or closure, effective sewage disposal may become worsen over time	Relevant stakeholders should be taken on board as soon as possible	PMU/ WASA, Lahore	Medium
		Change in government's priorities may affect sustainability of the project		Advocate for political commitment	PMU/ WASA, Lahore	
7	Temporal Delays	Delay during any phase/stage of the project cycle can causes multiple challenges	Delayed decisions or failure to meet project deadlines would have adverse affects	Push for timely decision making by all concerned authorities along with effective monitoring of given deadlines	PMU/ PMC/EPC Contractor	Medium
8	Execution Mode Risk	Lack of clarity in execution mode would delay in decision making and resultantly will impact the timely planning and execution of the project	Time delays would impact project cost	Clarity with respect to suitable execution mode is essential to investigate further risks and mitigation plans accordingly	PMU/ PMC/EPC Contractor	High

LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN E RAVI						
Sr. No	Risks	Causes/Factors	Effects	Risk Mitigation Plan	Institution/s Responsible to take corrective measures to mitigate Risk	Nature of Risk
9	Contract management of consultants / Responsibility of Risk	Lack of understanding/identification of risk ownership may result in multiple challenges during project execution	Primarily affects project execution and achieving desired objectives.	Legal and contractual risks should be clearly identified and properly transferred to relevant authorities/personnel/organization on at the contract finalization stage	PMU/ PMC/EPC Contractor	Low

12.6 PROCUREMENT PLAN:

The Standard Tender Document for Procurement of Plant Design, Supply, and Installation (AIIB Bank, May 2019) specifies the Lowest Evaluated Responsive Bid as the Contract Award Criteria and AIIB Standard Procurement Document [Large Works II] specify the Most Advantageous Bid as the Contract Award Criteria. Therefore, use of Standard Tender Document for Procurement of Plant Design, Supply, and Installation (AIIB Bank, May 2019) as the baseline document may result in a major shift in selection method of Contractors (Acceptance of bid and award of Contract). Hence single stage two envelope method has been proposed to be adopted for procurement followed by selection of contractors on EPC Mode. Relevant Email is attached at **Annexure-9**

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE									 WASA LAHORE DEVELOPMENT AUTHORITY					
Activity ID	Activity Name	BL Project Duration	BL Project Start	BL Project Finish	Original Duration	Start	Finish	Performance % Complete	Qtr 3, 2024				Qtr 4, 2024	
									Jun	Jul	Aug	Sep	Oct	Nov
 MTBM Laying of Sewerage System Through Tre														
 A1000	Updation and Submission of PC-1	60.00	01-Jun-24	10-Aug-24	60.00	01-Jun-24 A	10-Aug-24,	100%	 Updation and Submission of PC-1					
 A1010	Submission of Revised PC-1	60.00	01-Jun-24	10-Aug-24	60.00	01-Jun-24 A	10-Aug-24,	100%	 Submission of Revised PC-1					
 MTBM.2 Approval of PC-1														
 A1020	Pre-PDWP	10.00	15-Aug-24	27-Aug-24	10.00	15-Aug-24 A	27-Aug-24,	100%	 Pre-PDWP					
 A1030	PDWP	12.00	27-Aug-24	10-Sep-24	12.00	27-Aug-24 A	10-Sep-24,	100%	 PDWP					
 A1040	CDWP	50.00	12-Sep-24	09-Nov-24	50.00	12-Sep-24*	09-Nov-24	0%	 CDWP					
 A1050	ECNEC	30.00	09-Nov-24	14-Dec-24	30.00	09-Nov-24*	14-Dec-24	0%						
 MTBM.3 Negotiations with Bank														
 A1060	AIIB Negotiations with EAD	15.00	14-Dec-24	15-Jan-25	15.00	14-Dec-24	15-Jan-25	0%						
 A1070	Approval from AIIB	7.00	15-Jan-25	23-Jan-25	7.00	15-Jan-25	23-Jan-25	0%						
 MTBM.4 Tendering														
 A1080	Updation & Approval of Tender Documents	30.00	14-Dec-24	01-Feb-25	30.00	14-Dec-24	01-Feb-25	0%						
 A1090	Outreach Meetings	7.00	01-Feb-25	09-Feb-25	7.00	01-Feb-25	09-Feb-25	0%						
 A1100	Procurement Notice Preparation	7.00	10-Feb-25	18-Feb-25	7.00	10-Feb-25	18-Feb-25	0%						
 A1101	Approval & NOC from WASA & AIIB For Procuremer	5.00	18-Feb-25	23-Feb-25	5.00	18-Feb-25	23-Feb-25	0%						
 A1102	Procurement Notice of PPRA & Bank Website	7.00	23-Feb-25	02-Mar-25	7.00	23-Feb-25	02-Mar-25	0%						
 A1103	Procurement Notice in Newspapers	7.00	25-Feb-25	04-Mar-25	7.00	25-Feb-25	04-Mar-25	0%						
 A1104	Queries from Bidders	21.00	04-Mar-25	25-Mar-25	21.00	04-Mar-25	25-Mar-25	0%						
 A1105	Pre-Bid Meeting	3.00	25-Mar-25	28-Mar-25	3.00	25-Mar-25	28-Mar-25	0%						
 A1106	MOM Drafting	7.00	28-Mar-25	04-Apr-25	7.00	28-Mar-25	04-Apr-25	0%						
 A1107	MOM Approval from WASA & AIIB	7.00	04-Apr-25	11-Apr-25	7.00	04-Apr-25	11-Apr-25	0%						
 A1110	Site Visits	7.00	11-Apr-25	19-Apr-25	7.00	11-Apr-25	19-Apr-25	0%						
 A1120	Pre-Bid Meeting 2	3.00	19-Apr-25	23-Apr-25	3.00	19-Apr-25	23-Apr-25	0%						
 A1121	Drafting of Addendum 1	7.00	23-Apr-25	30-Apr-25	7.00	23-Apr-25	30-Apr-25	0%						
 A1126	Approval & NOC of Addendum 1 from WASA & Bar	7.00	30-Apr-25	07-May-25	7.00	30-Apr-25	07-May-25	0%						
 A1130	Submission of Tenders	75.00	07-May-25	02-Aug-25	75.00	07-May-25	02-Aug-25	0%						
 A1140	Evaluation of Technical Proposals	25.00	02-Aug-25	31-Aug-25	25.00	02-Aug-25	31-Aug-25	0%						
 A1150	NOC From AIIB & Announcement of Technical Evalt	10.00	01-Sep-25	12-Sep-25	10.00	01-Sep-25	12-Sep-25	0%						
 A1160	Opening of Financial Proposals	7.00	12-Sep-25	20-Sep-25	7.00	12-Sep-25	20-Sep-25	0%						
 A1170	Evaluation of Financial Proposals	14.00	20-Sep-25	07-Oct-25	14.00	20-Sep-25	07-Oct-25	0%						
 A1180	NOC from AIIB for announcement of Final Result	15.00	07-Oct-25	24-Oct-25	15.00	07-Oct-25	24-Oct-25	0%						
 A1190	Complain Redressal	15.00	24-Oct-25	11-Nov-25	15.00	24-Oct-25	11-Nov-25	0%						
 A1191	Negotiations With Successful Bidder	15.00	11-Nov-25	26-Nov-25	15.00	11-Nov-25	26-Nov-25	0%						
 A1192	Drafting of Work Contract	45.00	07-Oct-25	21-Nov-25	45.00	07-Oct-25	21-Nov-25	0%						
 A1193	Approval & NOC From WASA & AIIB For Work Contr	10.00	21-Nov-25	01-Dec-25	10.00	21-Nov-25	01-Dec-25	0%						
 A1200	Contract Award	7.00	01-Dec-25	09-Dec-25	7.00	01-Dec-25	09-Dec-25	0%						
 A1210	Submission of Performance Gaurantee	30.00	09-Dec-25	13-Jan-26	30.00	09-Dec-25	13-Jan-26	0%						
 A1220	Verification of PG	15.00	13-Jan-26	28-Jan-26	15.00	13-Jan-26	28-Jan-26	0%						
 A1230	Issuance of Notification of Commencement	7.00	28-Jan-26	04-Feb-26	7.00	28-Jan-26	04-Feb-26	0%						



13. MANAGEMENT STRUCTURE AND MANPOWER REQUIREMENTS

HR Management Plan

HR MANAGEMENT PLAN

Human Resource Management is the process of recruiting, selecting, inducting employees, providing orientation, imparting training and development, appraising the performance of employees, deciding compensation and providing benefits, motivating employees, maintaining proper relations with employees and their trade unions, ensuring employees safety, welfare and health measures in compliance with prevailing labor laws/service Rules.

Human Resource Management involves management functions like planning, organizing, directing and controlling

- It includes the recruitment, professional development, and management of human resources.
- It helps to achieve individual, organizational and social objectives
- Human Resource Management is a multidisciplinary subject. It includes the study of management, psychology, communication, economics and sociology.
- It involves team spirit, teamwork and it is a continuous process.

In order to fulfill HR requirements for instant PC-I, an independent Project Management Unit for potential project of LARECHS Trunk sewer has been proposed to be established for which Remuneration structure and JDs have already been approved by PDWP in its 15th Meeting held on 27.09.2022. PMU will work in close liaison with Project Management Consultant (PMC) and EPC Contractor regarding monitoring and implementation of project.

ROLE OF PROJECT MANAGEMENT UNIT

Brief responsibilities of PMU are mentioned hereunder:

- i. Responsible for overall delivery of the project.
- ii. Provide technical directions for the design & implementation of the project.
- iii. Overall financial and administrative management of the project as per rules of the Government and Guidelines of the funding agencies/banks.
- iv. Review and approval of detailed estimates and variations.
- v. Review and approval of documents / deliverables / payments as submitted by the Consultants and EPC Contractors.
- vi. Review / supervise the work of consultants and contractors.
- vii. Identify deficiencies in design, tender documents, contract agreement and provide effective solutions and timely redressal of issues.
- viii. Manage technical risks throughout the project. Supervise the progress of consultants & contractors / sub-contractors during project preparation & appraisal.

- ix.** Coordinate meetings and presentations to executing and funding agencies / banks and other government authorities as and when required.
- x.** Overall coordination with all stakeholders of the project including allied departments, affected persons and staff.
- xi.** Conduct site visits / inspections to finalize on-site issues / problems.
- xii.** Ensure work is being carried out as per approved standards and specifications.
- xiii.** Ensure key technical, economic, financial and social aspects are incorporated in the project and are dealt diligently and effectively.
- xiv.** Managing Team of engineers and allied personal, report writing & presentations and accomplishment of day-to-day activities / tasks.
- xv.** Participate in and contribute to relevant internal and external training and conferences and other activities for advocacy, dissemination and knowledge building.
- xvi.** Ensure the on-going learning and development of reporting staff.
- xvii.** Making payments to the consultants, contractors and allied departments and affected persons.
- xviii.** Responsible for the investigation and analysis of various incidents / troubles, and formulate optimization or rectification measures during the project.
- xix.** Evaluate the working efficiency of employees and propose adjustments.
- xx.** Communicate with employees, grasp the state of team atmosphere, and formulate measures to ensure the smooth operation.
- xxi.** Supervise the implementation of SOPs of Work.
- xxii.** Ensure project implementation schedule is followed evenly for timely completion of the project.
- xxiii.** Responsible for execution of works at site and management of the contract in entirety.
- xxiv.** Scrutinize claims of the contractors and consultants and give necessary approvals.
- xxv.** Bridge the coordination among contractor & consultants at each step.
- xxvi.** Provide all requisite updates for the Project Steering Committee of the project and liaison with the Bank.
- xxvii.** Budget Management in accordance with the provision of the loan amount and local component for the project.
- xxviii.** Responsible for implementation of procurement as per Procurement rules and guidelines of the MDBs / IFIs.

SIGNIFICANCE OF HIRING PMU:

The requirements of public administration institutions are rising and projects becoming increasingly challenging. Managing a project is a complex activity, particularly when it involves many people working over a long period of time in collaboration with many stakeholders. This increasing complexity requires management practices and tools, which ensure efficient use of resources. In this context, a PMU can be of great value as it will be provided with infrastructure, resources, processes, and tools necessary for effective project management by leveraging standards, allocating resources, and establishing communication channels.

PMU STRUCTURE **SEWERAGE SYSTEM FROM LARECHS COLONY TO GULSHAN E RAVI,** **LAHORE**

The PMU structure for the Trunk Sewer Project mentioned here under was approved by the PDWP in its 15th meeting held on 27.09.2022. Minutes of Meeting attached at **Annexure-10**. In the meeting Cost of PMU LARECHS was approved with the directions that administrative department will submit modified working papers to P&DD. Modified Working Paper sent to P&B board is appended at **Annexure-11**. Cost Breakdown of Revised Proposed PMU LARECHS in light of modified working Paper is also appended at **Annexure-12**.

Sr. #	Name of Post	Qty	Sr#	Name of Post	Qty
1	Project Director (PPS-12)	1	11	* Office & HR Manager (PPS-8)	1
2	TBM Expert / Geo-technical expert (PPS-12)	1	12	Assistant Manager (Legal) (PPS-8)	1
3	Manager (Civil) (PPS-10)	1	13	Accounts Assistant (PPS-5)	1
4	Manager (Electrical & Mechanical) (PPS-10)	1	14	Sub Engineer (1 Civil & 1 Electrical / Mechanical) (PPS-6)	2
5	Manager (Procurement & Contract) (PPS-10)	1	15	GIS Operator (PPS-5)	1
6	Manager (Environmental & Social) (PPS-9)	1	16	Office Assistant (PPS-3)	1
7	Manager (Admin & Finance) (PPS-9)	1	17	Vehicle Driver (PPS-2)	2
8	Assistant Manager (Civil) (PPS-8)	1	18	Naib Qasid (PPS-1)	2
9	Assistant Manager (Electrical) (PPS-8)	1	19	Security Guard (PPS-1)	2
10	Assistant Manager (Mechanical) (PPS-8)	1	20	Sweeper (PPS-1)	1

*Post of Office & HR manager is part of PMU as per requirement of the Bank and was part of final position paper & total budget approved from PDWP.

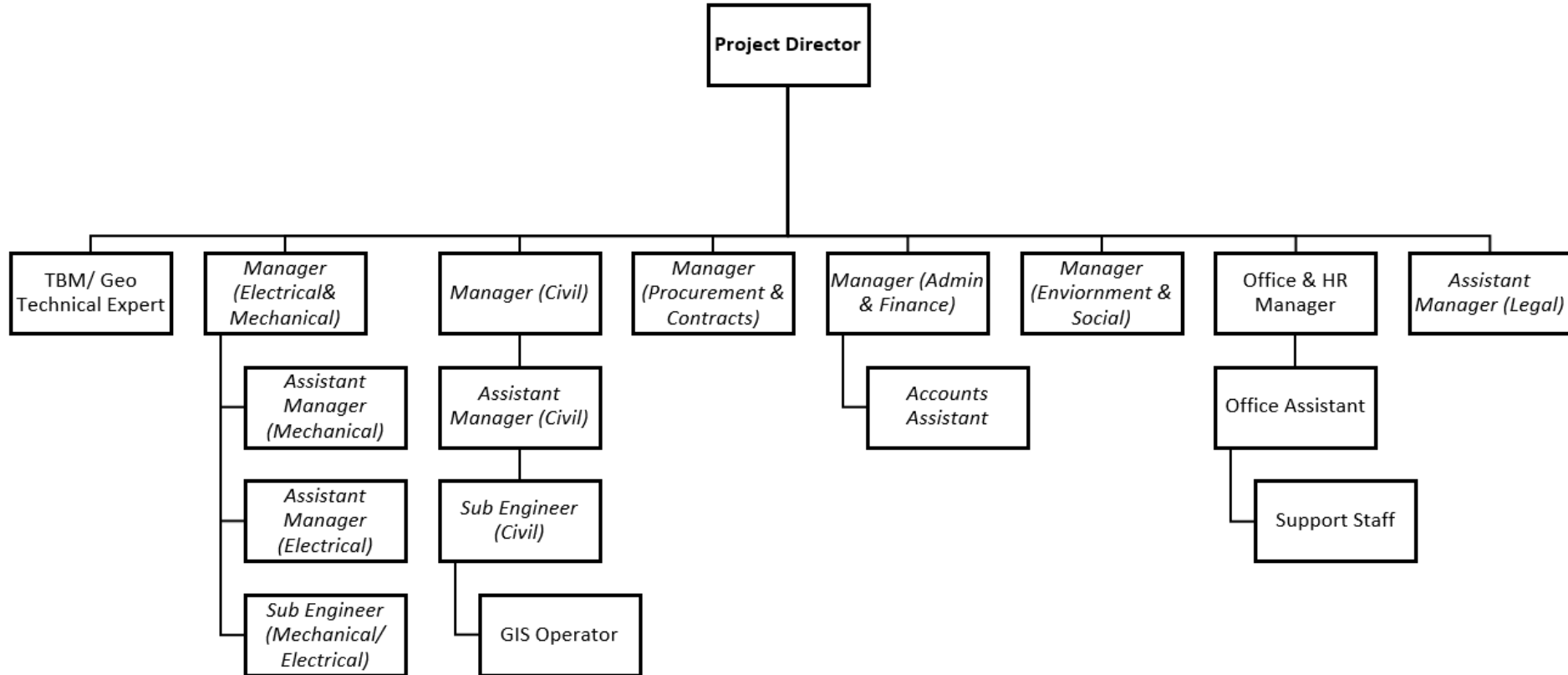


Figure 1: Project Management Unit (LARECHS)

It is to apprise that the matter regarding change in qualification and experience requirement of the TBM Expert / Geo-technical expert was placed before 5th Project steering Committee meeting held on 22.01.2024 under chairmanship of Chairman P&D board to relax the previously approved criteria of B.Sc. Civil Engineering or M.Sc. in Geotechnical Engineering to Bachelor/ Master degree in Geological/Mining Engineering or in relevant discipline to facilitate more applicants to apply for said post. In the meeting it was decided agenda regarding revision of qualification criteria of said post may be placed before relevant forum for consideration. Moreover, this post has been advertised both locally (multiple times) and internationally however, no suitable candidate has been shortlisted. Hence the new proposed qualification criteria with further relaxation as discussed during AIIB Mission dated 1-3 August 2024 is also been made part of instant revised PC-I for subsequent approval from PDWP forum and subsequently CDWP. (Minutes of 5th PSC attached at **Annexure-13**)

Position	Previously approved Qualification & Experience Criteria	Newly proposed Qualification & Experience Criteria
TBM Expert / Geo-Technical Expert (PPS-12)	B.Sc. in Civil Engineering (preferably with M.Sc. in Geotechnical Engineering) from HEC recognized university with a minimum of 15 years of relevant experience relating to micro-tunneling / tunneling. Experience related to Geotechnical investigations of laying sewer through micro-tunneling / tunneling of large-scale projects, especially in a metropolitan area will be highly preferred.	B.Sc. in Civil/ Geological/ Mining or relevant Engineering (preferably with M.Sc. in Geotechnical/Geological/Mining Engineering or in relevant Discipline) from HEC recognized university with a minimum of 08 years of relevant experience relating to micro-tunneling / tunneling. Experience related to Geotechnical investigations of laying sewer through micro-tunneling / tunneling of large-scale projects, especially in a metropolitan area will be highly preferred.

14. ADDITIONAL PROJECTS / DECISIONS REQUIRED

ADP Scheme "**Sewerage system from LARECH Colony to Gulshan-e-Ravi through Trenchless Technology**" was approved from Executive Committee of National Economic Council (ECNEC) for an amount of 14165.06 Million (Administrative approval was issued vide No SO(UD)3-4/2019 dated 25.06.2020, however, revision of instant same is being carried out owing to Upward revision in cost of Project Management Consultancy after completion of Procurement process and Change in Alignment of original route after detailed working by PMC based on prevailing site conditions, updated MRS rates and to account for all flows disposing into cantt drain.

Scheme ID	Scheme Name
01981812218	REVISED PC-1 OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE (THROUGH TRENCH-LESS TECHNOLOGY)

15. CERTIFICATE

Focal Person Name: PMU LWWMP

Email: syaa.engr@gmail.com

Fax No:

Address: WASA Head Office Zahoor Elahi Road Gulberg-II Lahore

Designation: PMU LARECHS

Tel. No.: 03333330492

PC-I FOR "SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E- RAVI
LAHORE (THROUGH TRENCHLESS TECHNOLOGY- 2nd REVISION)

Prepared By:

on Behalf of


PROJECT MANAGEMENT CONSULTANT (PMC)
EBA-ECSP-EEC-TEAM JV

Checked By:

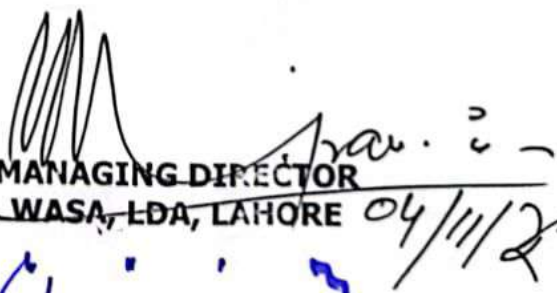

ASSISTANT PROJECT DIRECTOR
LWWMP,
WASA, LDA, LAHORE


MANAGER
PMU LARECHS

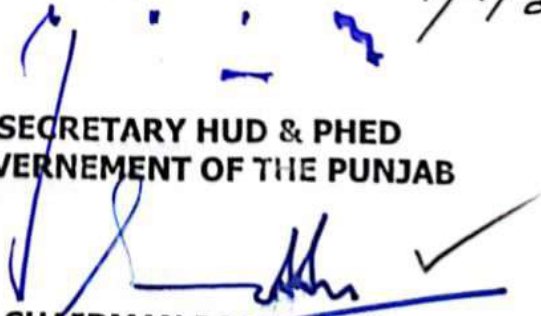

PROJECT DIRECTOR
LWWMP,
WASA, LDA, LAHORE

Forwarded By:


DY MANAGING DIRECTOR (ENGG.)
WASA, LDA, LAHORE


MANAGING DIRECTOR
WASA, LDA, LAHORE

SECRETARY HUD & PHED
GOVERNEMENT OF THE PUNJAB


CHAIRMAN P&D BOARD
GOVERNEMENT OF THE PUNJAB

**LAHORE WATER & WASTEWATER MANAGEMENT PROJECT
LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RABI, LAHORE
PACKAGE WISE PC-1 COMPARISON**

Sr. No.	Description	ORIGINAL APPROVED PC-1 AMOUNT (RS.) (June 2015)	1ST REVISION (June 2020)	2ND REVISION (June 2024)	DIFFERENCE	REMARKS
1	Package 1 - Laying of Sewer Line From Karachi Phatak and Gurumangat to Gulshan-E-Rabi	2,881,805,200 (Pkg-I)	4,888,408,627 (Pkg-I)	30,752,793,698	21,261,276,943	The upward increase is owing to following major factors: Machine Costs: 1. Addition of 02 No.s 1050 mm Machines (diff: 1,522 Billion) 2. Increase in cost of 1500mm machine. (diff: 315 Million) 3. Increase in cost of 2no. 2400mm machine instead of 1no. (diff: 1,716 Billion) 4. Addition of 3500mm machine in place of 2400mm. (diff: 2,199 Billion) Pipe Costs: 1. Addition of 1050 mm Pipe Length (1,746 Billion) 2. Increase in Cost of 1500mm. (diff: 7 Million) 3. Increase in Cost of 2400mm. (diff: 4,526 Billion) 4. Addition of 3500mm pipe length. (diff: 2,824 Billion) Shafts: 1. Change of design of pits (diff: - 593 Million) 2. Ventilation System & Screening System (Diff: 1,369 Billion) 3. Elimination of Lift Stations from 12 to 23no. (diff: 2,52 Billion) 4. SCADA System. (diff: 200 Million) 5. Addition of drop structures. (diff: 527 Million) 6. Sheet Piling & Bracing System. (Diff.: 1,816 Billion) Disposal Station: 1. Design change due to increase of capacity from 350 cfs to 688 cfs. (Diff.: 2,388 Billion) 2. Increase in pumps from 8 to 16no. (Diff.: 3,308 Billion) 3. Electrical System. (Diff. 813.2 Million) 4. Outfall Channel (Diff. 1,256 Billion) 5. Solar System (Diff.: 80 Million) Note: 1. MRS rates revised as per current Bi-Annual (From 1st Bi-annual 2019 to 2nd Bi-annual 2024) 2. Latest Currency Exchange rate applied (From 116 to 278.25) 3. MTBM Quantity (From 3 to 6) increased to complete project within actual timeline. 4. New MTBM considered instead of used machine
2	Package 2 - Disposal Station & Rehabilitation of Outfall Channel	700,000,000 (Pkg-II)	2,642,107,347 (Pkg-II)	9,065,661,849	6,423,554,502	Note: 1. MRS rates revised as per current Bi-Annual (From 1st Bi-annual 2019 to 2nd Bi-annual 2024) 2. Latest Currency Exchange rate applied (From 116 to 278.25) 3. MTBM Quantity (From 3 to 6) increased to complete project within actual timeline. 4. New MTBM considered instead of used machine
	TOTAL	6,085,937,553	12,133,624,101	39,818,455,547	27,684,831,446	
i)	Cost for Owner's Engineer Consultancy Services	194,750,002	354,721,596	1,274,380,658	919,659,062	The PMC Cost increased due to following reasons: 1. PMC Contract was awarded at a higher cost based on received Financial Bid for 707.43MPKR(1.6MUSD+401MPKR) as per recommendations of 3rd PSC date:24.01.2022. 2. To carryout detailed detailed geotechnical investigations as Amendment #1 for 65.2MPKR(0.081MUSD+42.3MPKR)based on recommendations of PMC the budget of which was not part of Bid. 3. Amendment # 2 for 126.7MPKR(0.283MUSD+47.2MPKR)for carrying out feasibility study of revised alignment based on recommendations of 5th PSC Meeting dated: 22.01.2024. 4. As per contract Inflation/escalation
ii)	ESMP including Livelihood Restoration Plan (LRP) & Resettlement plan, Health & Safety Cost	120,000,000	120,000,000	191,007,157	71,007,157	Will be charged as per actual.
iii)	Provision for Electrical works at Disposal Station	10,000,000	10,000,000	Included in Package - II (Sr. No. 2)		
iv)	Provision for Environmental Impact Assessment	2,000,000	2,000,000	Included in Separate PC-1		
v)	Provision for Electrical connecting charges at D/station/Grid Station		56,341,500	296,050,000	1,515,745,400	As Per current LESCO Rates however will be charged on actual LESCO estimates
vi)	PMU-L-WASA Establishment				239,708,500	
vii)	Shifting of Utility services @ 1 % of Project Cost	30,429,688	60,668,121	398,184,555	337,516,434	Will be charged as per actual.
viii)	Provision for Road Restoration / Remodelling & Bridge Construction @ 1 %	95,790,000	60,668,121	398,184,555	337,516,434	Will be charged as per actual.
ix)	Contingencies (excluding land acquisition and price escalation)	121,718,751	242,672,482	242,673,577	1,095	
x)		253,175,002			(2,000,000)	Will be charged as per actual.
xi)	Provision for Media Campaign @ 0.05 %	15,214,844	2,000,000		(121,336,241)	Will be charged as per actual.
xii)	Provision of Horticulture @ 0.05 %	60,859,376	121,336,241			
xiii)	Livelihood Restoration Plan		606,681,205	1,990,922,777	1,384,241,572	
xiv)	PRA @ 5%			1,860,620,000	487,809,279	
xv)	Duties & Taxes on Import Items			487,809,279		
xvi)	01 Years O&M		394,342,783	796,369,111	402,026,328	
xvii)	Price adjustment / Variation & Escalation @ 2 %					235%
	GRAND TOTAL	6,989,875,216	14,165,056,150	49,270,402,618	33,256,726,467	

On Behalf of

Assistant Project Director
LWWMP

Project Management Consultant (PMC)
EBA - ECSP - EEC - TEAM JV

Manager
PMU LARECHS

Project Director
LWWMP
WASA, LDA, Lahore

Dy. Managing Director (Engg.)
WASA, LDA, Lahore

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT
LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RAVI, LAHORE
ABSTRACT OF COST (PACKAGE WISE)

Exchange Rate

USD

EURO

3-May-24

278.25

300.56

SR. NO.	DESCRIPTION	AMOUNT
1	Package 1	
a	Sewer Line from Karachi Phatak to Qartaba Chowk	
b	Sewer Line from Gurumangat to Qartaba Chowk	8,918,781,370.54
c	Combined Reach (Toheed Park To Gulshan-e- Ravi)	7,069,358,999.55
d	Sheet Piling & Bracing System	8,277,031,033.36
e	Connections with Existing Disposal Stations to be eliminated	1,530,502,351.89
f	Inlet Screens at all Stations With Odor Control System, Drop Manholes	2,520,066,564.84
g	Rider Sewers & Bricklining of Cantt. & Gulberg Drains	1,369,346,954.65
h	Site Facilities	861,406,422.79
		206,300,000.00
	SUB TOTAL	30,752,793,698
	PRA @ 5% (Counterpart Funding)	1,537,639,685
	GRAND TOTAL	32,290,433,383
i	Shifting of Utility services @ 1 % of Project Cost	307,527,937
ii	Provision for Road Restoration / Remodelling & Bridge Construction @ 1 %	307,527,937
iii	Price adjustment / Variation & Escalation @ 2 %	615,055,874
	Total Amount (Rs.)	33,520,545,130
	Total Amount in US\$ (In Million) @ 1US\$ = Rs 278.25 (SBP Exchange Rate May 03, 2024)	\$ 120,469,165

SR. NO.	DESCRIPTION	AMOUNT
2	Package 2	
a	Construction of Disposal Station of 688 Cusecs Capacity at Gulshan-e-Ravi (Ultimate Capacity = 848 Cusecs)	8,503,792,970
b	Rehabilitation of Outfall Channel	395,218,880
d	Site Facilities	86,650,000
e	450 kW Solar System	80,000,000
	SUB TOTAL	9,065,661,849
	PRA @ 5% (Counterpart Funding)	453,283,092
	GRAND TOTAL	9,518,944,942
i	Shifting of Utility services @ 1 % of Project Cost	90,656,618
ii	Provision for Road Restoration / Remodelling & Bridge Construction @ 1 %	90,656,618
iii	Price adjustment / Variation & Escalation @ 2 %	181,313,237
iv	01 Years O&M	487,809,279
	Total Amount (Rs.)	10,369,380,695
	Total Amount in US\$ (In Million) @ 1US\$ = Rs 278.25 (SBP Exchange Rate May 03, 2024)	\$ 37,266,418

	GRAND TOTAL	AMOUNT
3		33,520,545,130
i	Package 1	10,369,380,695
ii	Package 2	1,515,745,400
iii	Provision for Electrical connecting charges at D/station/Grid Station	1,860,620,000
iv	Duties & Taxes on Import Items	1,274,380,658
v	Cost for Owner's Engineer Consultancy Services	242,673,577
vi	Contingencies (excluding land acquisition and price escalation)	191,007,157
vii	ESMP including Livelihood Restoration Plan (LRP) & Resettlement plan, Health & Safety Cost	296,050,000
viii	PMU-L-WASA Establishment	49,270,402,618
	GRAND TOTAL in PKR	49,270,402,618
	Total Amount in US\$ (In Million) @ 1US\$ = Rs 278.25 (SBP Exchange Rate May 03, 2024)	\$ 177,072,426

On Behalf of

Assistant Project Director
LWWMP
WASA, LDA, Lahore

Project Management Consultant (PMC)
EBA - ECSP - EEC - TEAM JV

Manager
PMU LARECHS

Project Director
LWWMP
WASA, LDA, Lahore

Dy. Managing Director (Engg.)
WASA, LDA, Lahore

LAHORE WATER & WASTEWATER MANAGEMENT PROJECT
LAYING OF SEWERAGE SYSTEM FROM LARECH COLONY TO GULSHAN-E-RABI, LAHORE
ABSTRACT OF COST

Exchange Rate
 USD
 EURO

3-May-24
 278.25
 300.56

Sr. No.	DESCRIPTION	AMOUNT (PKR)
A	Sewer Line from Karachi Phatak to Qartaba Chowk	8,918,781,370.54
B	Sewer Line from Gurumangat to Qartaba Chowk	7,069,358,999.55
C	Combined Reach (Toheed Park To Gulshan-e- Ravi)	8,277,031,033.36
D	Sheet Piling & Bracing System	1,530,502,351.89
E	Connections with Existing Disposal Stations to be eliminated	2,520,066,564.84
F	Construction of Disposal Station of 688 Cusecs Capacity at Gulshan-e-Ravi (Ultimate Capacity = 848 Cusecs)	8,503,792,969.66
G	Rehabilitation of Outfall Channel	395,218,879.65
H	Rider Sewers & Bricklining of Cantt. & Gulberg Drains	861,406,422.79
I	Inlet Screens at all Stations With Odor Control System, Drop Manholes	1,369,346,954.65
J	Site Facilities	292,950,000
K	Miscellaneous	
L	450 kW Solar System	80,000,000
	Sub-Total (A+B+C+D+E+F+G+H+I+J+K+L) (Rs.)	39,818,455,547
	TOTAL	39,818,455,547
	PRA @ 5% (Counterpart Funding)	1,990,922,777
	GRAND TOTAL	41,809,378,324
	Duties & Taxes on Import Items	1,860,620,000
i)	Cost for Owner's Engineer Consultancy Services	1,274,380,658
ii)	ESMP including Livelihood Restoration Plan (LRP) & Resettlement plan, Health & Safety Cost	191,007,157
iii)	Provision for Electrical connecting charges at D/station/Grid Station	1,515,745,400
iv)	PMU-L-WASA Establishment	296,050,000
v)	Shifting of Utility services @ 1 % of Project Cost	398,184,555
vi)	Provision for Road Restoration / Remodelling & Bridge Construction @ 1 %	398,184,555
vii)	Contingencies (excluding land acquisition and price escalation)	242,673,577
ix)	Provision for Media Campaign @ 0.05 %	
x)	Provision of Horticulture @ 0.05 %	
xi)	Price adjustment / Variation & Escalation @ 2 %	796,369,111
xii)	01 Years O&M	487,809,279
	Sub-total For Provisional Sums from item (i to xii)	7,461,024,293
	Total Amount (Rs.)	49,270,402,618
	Total Amount In US\$ (In Million) @ 1US\$ = Rs 278.25 (SBP Exchange Rate May 03, 2024)	177,072,426

[Handwritten Signature]